

Extending Intuitionistic Logic to a Probability Calculus (in Lean 4)

Anonymous Author(s)

Abstract

Kolmogorov's probability fixes the event algebra to be Boolean, so every event is decidable. This is a poor fit for the semidecidable properties, termination, reachability, that verification cares about. Following the Cox–Jaynes interpretation of probability as extended logic, we define a construct for extending a logic (treated as a parameter in our approach) to a calculus of probability, and specify criteria for when this works: (1) when the resulting calculus can reconstruct the original logic in the limiting case of certain (0– or 1– probability) events, and (2) when the construction, evaluated at classical logic, agrees with classical probability theory. Instantiating the construction at intuitionistic logic gives valuations on frames, complete Heyting algebras of open sets, the algebras of verifiable propositions. In this setting, we define slack, $1 - v(A) - v(\neg A)$, the classical complement rule's deficit and boundary mass in the measure model. Slack pinpoints exactly when Bayesian conditioning, which treats the evidence as fully decided, gives an incorrect update, disagreeing with Dempster's rule. A separate obstruction, a five-element counterexample, is why Cox's sum rule cannot be derived and must instead be posited. Classical probability credits the slack of a semidecidable property to its failure, an event no finite observation can confirm. The whole development is machine-checked in Lean 4 over mathlib, in roughly 5,000 lines with no sorry.

Keywords: intuitionistic logic, probability, Cox's theorem, belief functions, locales, Lean, mathlib, formalized mathematics

1 Introduction

Kolmogorov's probability [Kolmogorov 1933] assigns numerical values representing uncertainty to events. The event logic is classical, forming a Boolean σ -algebra. The assignments satisfy non-negativity, normalization, and σ -additivity. This commits the framework to the axioms of classical logic, including the law of excluded middle. Many properties of programs (e.g. "does this computation halt?", "does this process reach a given state?") are only *semidecidable*: a procedure confirms the property when it holds, and no procedure, in general, confirms its failure. The mismatch that results is not a contradiction. It is a failure of discrimination. A probability calculus built on a particular logic inherits that logic's identifications, so it cannot discriminate events the logic proves equal.

In the case of classical logic, this identification is the *law of excluded middle*: $\text{halts} \sqcup \neg \text{halts} = \top$, so the Boolean event algebra contains $\neg \text{halts}$. The complement rule $P(A^c) = 1 - P(A)$ then forces the remainder $1 - P(\text{halts})$ onto $\neg \text{halts}$, an event no finite observation can confirm. The events "not yet confirmed" and "refuted" are identified. For a program whose non-termination is independent of ZFC [Yedidia and Aaronson 2016], the classical value $P(\text{halts}) = 0$ is correct, yet neither running the program nor any ZFC proof ever establishes it. Uniformly over programs, computing these sharp values is exactly as hard as the halting problem.

Another failure of discrimination appears in *linear logic*, whose tensor $\otimes$ distinguishes a from $a \otimes a$, one use of a resource from two. Modularity with $\otimes$ in place of $\sqcup$, $v(a) + v(b) = v(a \sqcup b) + v(a \otimes b)$, inherits idempotence of disjunction, $a \sqcup a = a$, and at $b = a$ forces $v(a \otimes a) = v(a)$. So, no assignment satisfying this axiom distinguishes one use of a from two.

Probability is, as the Cox–Jaynes tradition has it, *extended logic* [Cox 1946; Jaynes 2003], and different logics can be extended through the same construction. The logic of events, in its algebraic form the event algebra, is a *parameter* of the construction, and the choice of event algebra determines the probability calculus. An event algebra consists of a set of events together with operations interpreting the logic's connectives and structural rules, while a probability calculus is differentiated from an arbitrary grading of events by satisfying certain axioms. We specify how to construct such extensions, and propose two criteria for when the extension of a logic is a probability calculus:

- (1) the valuations taking only the values 0 and 1 (the two degrees of complete certainty) must reconstruct the underlying event logic. This is what we take "extended logic" to mean.
- (2) the construction evaluated at classical logic must return the classical theory, ensuring it is a genuine generalization of classical probability.

We propose two fixes for the identifications above. Both extend a logic to a probability calculus, and on finite frames the second reduces to the first. The first fix we propose is one that accepts the standard axioms imposed on a grading to make it a probability calculus, while choosing an event algebra that makes the required distinctions. The extension of the event algebra to a probability calculus is then given by the *valuations*, i.e. the maps from events to numbers satisfying monotonicity, normalization, and modularity. The

resulting extension does not guarantee defining a probability calculus, and we specify the conditions under which it does: the $\{0, 1\}$ -valued valuations are the points of the event algebra, and they reconstruct its order when it is distributive. Intuitionistic logic, in particular, is a case for which this approach yields an extension, and we work it out in detail. Its event algebras are the Heyting algebras, and we work with frames (the complete Heyting algebras). The open sets of any topological space form a frame. On a frame the pseudocomplement $\neg a$ need not satisfy $a \sqcup \neg a = \top$, so the complement rule weakens to $v(a) + v(\neg a) \leq 1$. The gap $\text{slack}(a) = 1 - v(a) - v(\neg a)$ is the mass of the part of a neither confirmed nor refuted. For the valuation that a measure μ induces on the open sets of a space, the slack of an open set is the μ -measure of its boundary.

The second fix also applies to logics whose connectives are not lattice operations, such as linear logic. It generalizes the axioms of the probability calculus in addition to varying the event algebra. It applies to any logic presented by a set of *certain states*, its $\{0, 1\}$ -valued assignments. The calculus is the set of finite mixtures of certain states. A mixture that takes only the values 0 and 1 is one of the certain states it mixes, because every certain state with positive weight must agree with it at every event. So the $\{0, 1\}$ -valued elements of the calculus are exactly the certain states, which present the logic, and criterion (1) holds. For finitely many events, the axioms of the calculus are exactly the affine laws that hold on every certain state. On finite frames and finite Boolean algebras these laws coincide with monotonicity, normalization, and modularity, and the calculus is the set of valuations.

For linear logic, the event algebra is a quantale, a complete lattice with an associative tensor $\otimes$ that need not be idempotent. The first fix fails there because of the probability axioms, not the event algebra: the modularity axiom identifies a with $a \otimes a$. Probabilistic coherence spaces [Danos and Ehrhard 2011] avoid this by assigning weights to the points of each formula's web instead of to formulas, with $a \otimes a$ interpreted on a product web. Their elements are not valuations in the sense of the modularity axiom. For every multiplicative-additive formula over finite data types, the calculus of the second fix is contained in the probabilistic coherence space of the formula, and the certain elements of that space are cliques of Girard's coherence space [Girard 1987].

Our extension and coherence spaces coincide at first-order linear logic. At higher order they can differ: for a program with two functional arguments the calculus is the set of randomized sequential programs, a strict subset of the probabilistic coherence space (Section 11). The probabilistic coherence space, the standard model of probabilistic programs at this type, therefore admits behaviours that no sequential program flipping coins can produce, if programs are modelled as behavioural strategies. Reasoning in it can certify only weaker guarantees than the programs satisfy. The gap has

the same numerical form as the gap between classical and non-classical correlations in quantum foundations [Cabello et al. 2014]. At !Bool the calculus is also a strict subset.

Quantum logic, whose event algebras are orthomodular lattices [Birkhoff and von Neumann 1936], is a logic to which neither fix applies. Both rely on $\{0, 1\}$ -valued assignments: the first on $\{0, 1\}$ -valued valuations, which do not exist on non-distributive lattices, and the second relies on having certain states. A certain state of quantum logic would assign 1 to exactly one ray of every orthogonal basis, and by the Kochen–Specker theorem [Kochen and Specker 1967] there is none in dimension at least 3. We prove this for an 18-ray configuration in $\mathbb{R}^4$ [Cabello et al. 1996], so the calculus of the second fix is empty there. Quantum states, which by Gleason's theorem [Gleason 1957] are given by density matrices, nevertheless exist, and they fail criterion (1).

Contributions.

1. **Weak excluded middle** (Sections 2 and 7). Global weak excluded middle on the opens of X is exactly extremal disconnectedness of X , for compact Hausdorff X exactly the Stonean spaces.
2. **Two characterizations** (Sections 2, 3 and 5). The complement rule holds for *every* valuation iff the frame satisfies excluded middle, the exact content of Van Horn's classical negation axiom R3 [Horn 2003]. Dempster's rule and Bayesian conditioning, classically identical, agree iff the evidence has zero slack. The conditional $v(a \sqcap b)/v(b)$, the product rule, and Bayes symmetry hold unchanged. The complement rule weakens to the interval $[v(A), 1 - v(\neg A)]$ of width $\text{slack}(A)$, and the law of total probability to a lower bound whose deficit is a conditional slack.
3. **Structure theorems** (Sections 4 and 5). On finite frames a valuation is a classical mass function on points. In general it splits into atomic and diffuse parts, and on its Boolean core it is a belief function.
4. **Cox's theorem, dissected** (Section 6, Appendix C). Cox's theorem and its repairs [Cox 1946; Halpern 1999; Horn 2003] derive classical probability from qualitative requirements on plausibility. The product-rule half is independent of the logic, proved by an Aczél/Hölder generator built from scratch. The sum-rule half, a theorem classically, admits no derivation here: on a five-element frame a disjunction's value is not a function of its disjuncts' values, so modularity must be posited. Weak excluded middle does not restore it, since the counterexample frame satisfies weak excluded middle.
5. **Two grounding models** (Section 7). Any classical measure, read on the opens, is a valuation whose slack is boundary mass. A sharp readout of halting is uncomputable, and on Cantor space the value of the halting event is Chaitin's Ω . A classical posterior overstates

refutation by exactly the conditional boundary mass $\mu(\partial A \cap B)/\mu(B)$ (Theorem 33), so the classical and intuitionistic readings differ only where boundary mass is positive. Slack is submodular yet does not determine the valuation.

6. **The monad dichotomy** (Section 8). Product valuations are unique on $\text{Low}(\mathbb{N} \times \mathbb{N})$ and provably non-unique on $\text{Set}(\mathbb{N} \times \mathbb{N})$, where Fubini fails. The countably-presented fragment composes associatively and commutatively.
7. **Beyond frames** (Section 9). The certain valuations reconstruct the event order exactly on distributive lattices, and completeness is not needed. On the non-distributive diamond M_3 no certain valuation exists, so the construction does not extend to quantum event lattices unchanged. Grading linear logic's $\otimes$ by modularity forces $v(a \otimes a) = v(a)$, so such a grading cannot count resources.
8. **Probability calculi from certain states** (Section 10). The calculus of a logic presented by its $\{0, 1\}$ -valued certain states is the set of their finite mixtures. Its certain elements are exactly the certain states, and its axioms are exactly the affine laws valid on the certain states. On finite frames this returns the valuations, and on finite Boolean algebras Kolmogorov's calculus. For arbitrary sets of events, the functions satisfying every finitely supported law valid on the certain states are exactly the barycenters of probability measures concentrated on the certain states.
9. **Linear logic** (Sections 10.2 and 10.3). Every multiplicative-additive formula over finite data types denotes a probabilistic coherence space, whose certain elements are cliques of Girard's coherence space, and the calculus of the formula is contained in it. At implications between data types the two coincide. At $!\text{Bool}$ the calculus is a strict subset.
10. **Second-order linear types** (Section 11). For a program with two functional arguments over finite data types, the certain elements of the probabilistic coherence space are the deterministic sequential programs, and the calculus is exactly the set of randomized sequential programs, by both directions of Kuhn's theorem [Kuhn 1953]. The calculus is a strict subset of the probabilistic coherence space. The space is the clique-constrained polytope of an exclusivity graph whose stable-set polytope is the calculus, so the two can agree only if the graph has no odd hole or anti-hole. One tensor further, a clique of Girard's coherence space, Shannon's stable set of $C_5 \boxtimes C_5$, is not an element of the probabilistic coherence space.
11. **Quantum event logic** (Section 12). For the 18 rays and 9 orthogonal bases of Cabello's Kochen–Specker set in $\mathbb{R}^4$ [Cabello et al. 1996], no $\{0, 1\}$ -valued assignment gives exactly one ray of every basis the value

1, so the calculus is empty, while every Born-rule assignment of a unit vector sums to 1 on every basis. Quantum probability on this event logic fails criterion (1).

12. **A first formal treatment.** Everything above is machine-checked in Lean 4 over mathlib, to our knowledge the first mechanization of non-additive probability over intuitionistic logic.

Mechanization. The development is carried out in Lean 4 [de Moura and Ullrich 2021] over mathlib [mathlib Community 2020], in roughly 5,000 lines with no sorry, and is included in full as anonymous supplementary material.¹ Two points are specific to a proof assistant. We separate *object-level* classicality, excluded middle in the frame Ω , from the *meta-level* classicality mathlib inherits through `Classical.choice`, identify the four places where the meta-level dependence is essential (Section 3), and give choice-free proofs of the finite separation results. The product-rule half of Cox's theorem needs an Aczél/Hölder regreduation absent from mathlib, which we build from scratch.

2 Valuations on Frames

2.1 The Event Logic

Definition 1 (frame). A *frame* is a complete lattice Ω in which finite meets distribute over arbitrary joins,

$$a \sqcap \left(\bigsqcup_{i \in I} b_i \right) = \bigsqcup_{i \in I} (a \sqcap b_i)$$

for every $a \in \Omega$ and every family $\{b_i\}_{i \in I} \subseteq \Omega$. Equivalently, it is a complete Heyting algebra. By $x \Rightarrow y$ we denote the Heyting implication in a frame, and $\neg a := a \Rightarrow \perp$ is the pseudocomplement of $a \in \Omega$. Also, a^c denotes a complement of a , if it exists, i.e. an element such that $a \sqcap a^c = \perp$ and $a \sqcup a^c = \top$.

Frames are the algebraic semantics of intuitionistic logic and the point-free abstraction of topological spaces. The same objects with morphisms reversed form the category of *locales*, so the distinction is invisible at the level of a single algebra. We say *frame* throughout and reserve *locale* for the spatial and categorical statements (points of a locale, classifying locales) where that word is the standard one. The motivating example throughout is $\Omega = \mathcal{O}(X)$, the opens of a space, with $\sqcap, \sqcup$ as intersection and union and the pseudocomplement $\neg a$ of an open given by the *interior* of its set complement. In mathlib this is the typeclass `Order.Frame`, and the Heyting operations come with the key asymmetry already in place: $a \sqcap \neg a = \perp$ holds in any Heyting algebra (`inf_compl_eq_bot`), while $a \sqcup \neg a = \top$ is available only in

¹Claude (Anthropic) was used to assist with drafting and debugging the Lean formalization.

Boolean algebras. An element with $a \sqcup \neg a = \top$ is *complemented*: it satisfies its own instance of excluded middle.

2.2 The Valuation Structure

The central object assigns a degree of certainty to each element of the frame, with inclusion–exclusion as the only interaction axiom:

Definition 2 (valuation^{Lean}). A *valuation* on a frame Ω is a *monotone* function $v : \Omega \rightarrow [0, \infty]$ with $v(\perp) = 0$, $v(\top) = 1$ (*normalization*), and, for all $a, b \in \Omega$, *modularity*:

$$v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b).$$

```

structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b
    = toFun (a ⊔ b) + toFun (a ⊓ b)

```

This axiom set is not ours: Weatherson’s axioms for intuitionistic probability are exactly these [Weatherson 2003], and the valuations of constructive measure theory [Coquand and Spitters 2009; Vickers 2008] are their Scott-continuous instances (Section 13).

Values live in the extended non-negative reals $[0, \infty]$, and normalization confines them to $[0, 1]^{\text{Lean}}$. The conventions this choice carries, unconditional summation and truncated subtraction, are an artifact of the mechanization and are recorded in Appendix A.

2.3 Slack

Modularity applied to the disjoint pair $(a, \neg a)$ collapses to $v(a) + v(\neg a) = v(a \sqcup \neg a)^{\text{Lean}}$. So, the join need not be $\top$:

Theorem 3 (the complement bound^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $v(a) + v(\neg a) \leq 1$.*

```

theorem Valuation.add_compl_le_one (v :
  Valuation Ω) (a : Ω) :
  v a + v ac ≤ 1

```

This is a *derived* Dempster–Shafer sub-additivity inequality. The deficit

$$\text{slack}(a) = 1 - (v(a) + v(\neg a)) = 1 - v(a \sqcup \neg a)$$

is the valuation of the undecided region. It decomposes canonically. The pair $(\neg a, \neg a)$, double negation against negation, is always disjoint. However, it is a partition only under *weak* excluded middle ($\neg a \sqcup \neg \neg a = \top$). Applying modularity to this disjoint pair splits the slack into two logically independent gaps:

Definition 4 (regular element). An element $a \in \Omega$ is *regular* (or $\neg\neg$ -stable) if $a = \neg\neg a$.

Theorem 5 (slack decomposition^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{slack}(a) = \text{dnGap}(a) + \text{deMorganGap}(a)$, where $\text{dnGap}(a) = v(\neg\neg a) - v(a)$ is 0 if a is $\neg\neg$ -stable (regular), and $\text{deMorganGap}(a) = 1 - v(\neg\neg a \sqcup \neg a)$ is 0 if weak excluded middle holds at a .*

The single Dempster–Shafer “ignorance” number is thus resolved into a regularity defect and a De Morgan defect: the first measures failure of double-negation stability, the second corresponds to a named intermediate logic (weak excluded middle). Constructively, the contrapositive of a universal hypothesis returns only its double-negated form, so for open sets, this specification of probability does not fall victim to Hempel’s raven paradox [Hempel 1945]: evidence of the white sock indoors confirms at most $\neg\neg a$, not a itself, and dnGap is exactly the gap between confirming $\neg\neg a$ and confirming a itself.

Because dnGap and deMorganGap are themselves defined from v , the apparent three unknowns $v(a)$, $v(\neg a)$, and $v(\neg\neg a)$ are not independent: they are part of a three-constraint system, yielding the following result:

Theorem 6 (slack in closed form^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$,*

$$\text{slack}(a) = (v(\neg\neg a) - v(a)) + (1 - v(\neg\neg a) - v(\neg a)),$$

where $v(a) \leq v(\neg\neg a)$.

The inequality follows from monotonicity applied to $a \leq \neg\neg a$, a fact of every Heyting algebra, and shows the first summand is exactly the difference, not the floor produced by $[0, \infty]$ ’s truncated subtraction. The split into a regularity defect and a De Morgan defect follows from monotonicity and modularity alone, with no further axiom about ignorance required. This contrasts with the usual Dempster–Shafer picture, where the mass assigned to non-singleton, ambiguous sets is an extra primitive.

Theorem 6 expresses slack pointwise in terms of v . The converse, whether slack in turn determines v , is more delicate. One constraint is automatic:

Theorem 7 (slack is submodular^{Lean}). *For every valuation v on a frame Ω and every $a, b \in \Omega$,*

$$\text{slack}(a \sqcup b) + \text{slack}(a \sqcap b) \leq \text{slack}(a) + \text{slack}(b).$$

The proof runs v ’s own modularity twice, once at (a, b) and once at $(\neg a, \neg b)$, and combines the two results with the De Morgan laws: the equality $\neg(a \sqcup b) = \neg a \sqcap \neg b$, which holds in every Heyting algebra, and the inequality $\neg a \sqcup \neg b \leq \neg(a \sqcap b)$, monotone under v but not in general an equality. Any function proposed as the slack of some valuation must therefore be submodular.

Remark 8 (non-recovery^{Lean}). Submodularity is necessary, not sufficient: Section 7.2 exhibits two valuations on the same frame with identical slack functions.

Remark 9 (slack under weak excluded middle^{Lean}). The two gaps, dnGap and deMorganGap , computed at a single element, are logically independent. However, if weak excluded middle is imposed on the frame for every a ($\neg\neg a \sqcup \neg a = \top^{\text{Lean}}$), then deMorganGap vanishes identically and $\text{slack}(a) = \text{dnGap}(a)$ for every valuation and every a . This sits strictly between plain intuitionistic logic, where both gaps can be positive, and full excluded middle, where both vanish^{Lean}. A concrete instance appears in Section 7.

2.4 What Survives without Excluded Middle

Three classical pillars survive intact, and one fails in a precisely delimited way. *Disjoint additivity* is unconditional: $a \sqcap b = \perp$ implies $v(a \sqcup b) = v(a) + v(b)$. *Conditioning* is native: for $v(b) \neq 0$ the posterior $v(a \mid b) = v(a \sqcap b)/v(b)$ is again a valuation (the proof is frame distributivity), and the product rule $v(a \mid b) \cdot v(b) = v(a \sqcap b)$ and Bayes symmetry (and therefore Bayesian updating) $v(a \mid b) v(b) = v(b \mid a) v(a)$ hold^{Lean}. *Total probability* holds over any genuine partition, meaning $a \sqcap a' = \perp$ and $a \sqcup a' = \top$: $v(b) = v(a \sqcap b) + v(a' \sqcap b)$. What fails is total probability over $\{a, \neg a\}$. That family need not join to $\top$, and the conditional masses fall short of $v(b)$ by a conditional slack^{Lean}.

The classical fragment. We characterize the elements of Ω on which the classical complement rule holds. The $\neg\neg$ -stable (*regular*) elements, which form a Boolean algebra (by Glivenko's theorem), are a natural candidate, but they do not suffice. Modularity constrains the frame join $\sqcup$, whereas the Boolean join of regular elements is the regularized $\neg\neg(a \sqcup b)$, for which only the inequality $v(a) + v(b) \leq v(\neg\neg(a \sqcup b)) + v(a \sqcap b)$ survives (i.e. the 2-monotonicity of Section 5). A regular but uncomplemented element, which is an open ray $(-\infty, 0)$ in the opens of $\mathbb{R}$, under an atomic measure at 0, still carries slack. The correct classical fragment is the *complemented* elements, where $v(a) + v(a^c) = 1$ holds exactly^{Lean}. On a connected locale that fragment can be as small as $\{\perp, \top\}$. In the Boolean limit, every element is complemented, the slack vanishes identically, and classical Kolmogorov axioms are recovered^{Lean}.

Mixtures. Valuations are closed under finite convex combination^{Lean}, with every axiom inherited termwise, because modularity is linear. This innocuous fact becomes a characterization in Section 4.

3 Classicality in Theory and Metatheory

Classicality enters this work twice: in the metatheory, the logic of the proof assistant itself, and in the object theory, the frame's own logic.

The metatheory. Lean's kernel is constructive, and its core library declares `Classical.choice` as an axiom, from which excluded middle follows. A theorem is constructive when its proof does not use this axiom, which `#print` axioms reports. `mathlib` uses the axiom throughout. The order on $\mathbb{R}$ and $[0, \infty]$ has no computable Decidable instance, so every library lemma that case-splits on $x \leq y$ for such numbers uses `Classical.propDecidable`, and so do the library's finite sums and finite sets. A theorem that never invokes excluded middle for frame elements $a \in \Omega$ can therefore still depend on the axiom through the library. Almost every theorem of this development does. This dependence is incidental: it comes from the library, not from the argument. Table 1 lists the classical principles the arguments themselves use, and what a constructive version would require.

The finite combinatorial results (M_3 , the pentagon, the perfect-graph criterion, the Kochen–Specker set, and the multiplicative–additive formulas) need none of these principles. The Cox model of Section 6 is given an explicit computable Decidable instance.

Object-level classicality: the R3 axiom. Van Horn's repaired Cox theorem [Horn 2003] axiomatizes plausibility over classical propositions. Its negation axiom, called *R3*, posits that the plausibility of $\neg A$ is a fixed non-increasing function S of the plausibility of A . Consistency forces the involution $S \circ S = \text{id}$, and therefore the complement rule $p \mapsto 1 - p$. We show that the involutive negation is double-negation elimination, so *R3* is where classical logic enters every known Cox-style derivation. To do this, we first isolate the exact content of *R3*'s conclusion as a predicate on valuations, then prove equivalence to excluded middle.

That predicate is *classical negation*: a valuation v has classical negation if $v(a) + v(\neg a) = 1$ for every $a \in \Omega$.

Theorem 10 (the *R3* characterization^{Lean}). *For every frame Ω , every valuation on Ω has a classical negation iff $a \sqcup \neg a = \top$ for every $a \in \Omega$.*

The forward direction is a calculation. The content is the converse: wherever excluded middle fails, a valuation with genuine slack must be *constructed*. Suppose $a \sqcup \neg a \neq \top$.

Definition 11 (ideal, prime ideal). An *ideal* $J \subseteq \Omega$ is a downward closed set which is also closed under finite joins. An ideal J is *prime* if $\top \notin J$ and $a \sqcap b \in J$ implies $a \in J$ or $b \in J$.

The prime ideal separation theorem (`mathlib's` `DistribLattice.prime_ideal_of_disjoint_filter_ideal`) yields a prime ideal J containing $a \sqcup \neg a$ but not $\top$. The indicator of J 's complement,

$$v(x) = \begin{cases} 0 & x \in J \\ 1 & x \notin J, \end{cases}$$

Result	Classical principle	Constructive version
Converse of Theorem 10, separation (Sec. 9)	prime ideal theorem	finite lattices: proved without choice ^{Lean} ; infinite frames: only for frames with enough points
Completeness (Thm. 63)	Hahn–Banach over $\mathbb{R}$	for rational points and integer certain states, proved without choice by Fourier–Motzkin elimination over $\mathbb{Z}^{\text{Lean}}$
Barycenters (Thm. 65)	compactness of probability measures	finite E : the completeness row; in general the fan theorem is needed
Sharp readout (Sec. 7.3)	excluded middle for halting	not definable; replaced by a synthetic undecidability statement [Forster et al. 2019]
Non-uniqueness of products (Thm. 48)	free ultrafilter on $\mathbb{N}$	not on $\text{Set}(\mathbb{N} \times \mathbb{N})$; proved without choice on the algebra of sets eventually determined by the order of the coordinates ^{Lean}
Sierpiński model (Sec. 7.3)	bivalence of propositions	an explicit two-point space, proved without choice ^{Lean}
Slack is boundary mass (Thm. 32)	classical measures and set complements	finite spaces, proved without choice ^{Lean} ; in general constructive valuations [Coquand and Spitters 2009]
Kraft bound for Ω (Sec. 7.3)	product measure on Cantor space	finite Kraft inequality, proved without choice ^{Lean} ; Ω is the supremum of its partial sums, a lower real
Stonean characterization (Sec. 7.1)	Stone duality for spaces	none; the localic statement is the definition
Values in $[0, \infty]$, infinite sums	classical extended reals	rational numbers for finite results; lower reals in general, in which slack is an upper real
Aczél regraduation (Sec. 6)	classical real analysis	not attempted

Table 1. Classical principles used by the arguments. “Proved without choice” means that `#print axioms` lists at most `propext` and `Quot.sound`. These proofs use core Lean and, for the Fourier–Motzkin proof, `mathlib’s ring tactic` on $\mathbb{Z}$, and avoid `mathlib’s` $\mathbb{Q}$ and finite sums, which depend on `Classical.choice`.

is a valuation. Modularity in the only nontrivial case ($x, y \notin J$) is *exactly* primeness, $x \sqcap y \in J \Rightarrow x \in J \vee y \in J$. And this valuation assigns $v(a) + v(\neg a) = 0 \neq 1$.

Sharp valuations are the points of the locale. The construction above is one half of an exact correspondence. Call v *sharp* if it is $\{0, 1\}$ -valued, a state of full certainty. Sharpness characterizes exactly the prime-ideal indicators:

Theorem 12 (sharp valuations are points^{Lean}). *For every valuation v on a frame Ω , v is sharp iff v is the complement-indicator of a prime ideal.*

The correspondence is given by two mutually inverse maps: $v \mapsto Z(v) = \{x \in \Omega : v(x) = 0\}$, which computes the zero-set of a sharp valuation, and the map taking J to its complement-indicator (the valuation of Theorem 10’s construction). $Z(v)$ is prime exactly because v is sharp, and the complement-indicator of $Z(v)$ recovers v itself. In locale-theoretic terms, sharp valuations recover the *points* of the locale. A prime ideal is *completely prime* when it is also closed under arbitrary (not just finite) suprema of its members. The finitely prime points of Theorem 12 sharpen to the completely prime points exactly where Scott continuity is added, recovering points of the locale in the sense used by [Coquand and Spitters 2009; Vickers 2008].

Definition 13 (directed set, Scott continuity). A subset $D \subseteq \Omega$ is *directed* if, for any finite subset $F \subseteq D$, there exists an element $d \in D$ that is an upper bound of F (taking $F = \emptyset$ shows a directed set is nonempty). A valuation v is *Scott-continuous* if $v(\bigsqcup D) = \sup_{a \in D} v(a)$ for every directed D .

Theorem 14 (Scott-continuous sharp valuations^{Lean}). *For every valuation v on a frame Ω , v is sharp and Scott-continuous iff v is the complement-indicator of a completely prime ideal.*

The characterization and the two correspondences together locate the object-level classicality precisely: whenever excluded middle fails at some a , the failure occurs at a point of the locale.

Finite frames without choice. The prime ideal theorem is a choice principle. Without it, a frame can have no points at all [Johnstone 1982], and then it has no sharp valuations, so the converse of Theorem 10 and the statement that sharp valuations reconstruct the logic (the first criterion of Section 1) hold only in a classical metatheory. On a finite distributive lattice with decidable order the needed prime filter is found by search: an element below a and not below b that is minimal for the size of its downset is join-prime, and the indicator of its principal filter is a sharp valuation. This proves the finite separation theorem and the finite form of the converse of Theorem 10 without `Classical.choiceLeanLean`, for $\mathbb{N}$ -valued valuations. In a constructive metatheory the first criterion therefore holds for finite event algebras and for frames with enough points, and not in general.

A development in Rocq. Rocq’s kernel, like Lean’s, does not assume excluded middle or choice, so the two differ in their libraries more than in their logic. The differences would be the following.

- *Finite results.* The Mathematical Components library [Mahboubi and Tassi 2021] works with types with decidable equality and finite types, and avoids classical axioms in its algebraic core. The finite theorems of this paper (the finite representation, the frame recipe, M_3 , the pentagon, the perfect-graph criterion, and the Kochen–Specker set) would be proved there without classical axioms and could be extracted to programs.
- *Values.* $[0, \infty]$ is a classical object. A constructive development would use rational values for finite results and lower reals, approximable from below, for valuations in general [Coquand and Spitters 2009; Vickers 2008]. The value of the halting event, Chaitin’s Ω , is a lower real of this kind: it is approximated from below by the halting programs found so far, and no better.
- *Analysis.* MathComp-Analysis [Affeldt et al. 2018] assumes classical axioms, as mathlib does. The completeness theorem for rational points is the Fourier–Motzkin argument above, which carries over directly. Completeness over real points would require the constructive analysis of Bishop’s school, with locatedness hypotheses for hyperplane separation. The barycenter theorem for infinitely many events requires compactness of $\{0, 1\}^{\mathbb{N}}$, which constructively is the fan theorem and is not provable in Rocq’s logic without an additional axiom.
- *Computability.* Without excluded middle the sharp classical readout cannot be defined. The uncomputability statement would be phrased synthetically [Forster et al. 2019]: no function from codes to Booleans decides halting, which requires an axiom such as Church’s thesis to state as a negative result about all functions.
- *Points.* The separation results would be stated for frames with enough points, or for finite lattices as above.

Lean was chosen for mathlib’s measure theory, order theory and topology, which the infinite results use. The cost is that constructive content is a property to be checked theorem by theorem with `#print axioms`, rather than a default.

A natural direction for future work is to develop the extension of logics to probability calculi in a constructive metatheory throughout. This would give algorithms in place of existence proofs, for instance the weights of a convex combination and the separating laws of Theorem 63, and would match the semidecidable reading of events with value types such as lower reals. It would also restrict the results that depend on points existing to frames with enough points.

4 Valuation Representation

Every valuation on a finite frame is a probability mass function on points. On infinite frames, this can fail, and Scott

continuity marks the boundary between where it still holds and where it does not.

4.1 Finite Frames Representation

Theorem 15 (Birkhoff duality [Birkhoff 1937]). *Every finite distributive lattice is isomorphic to the lattice of lower sets of a finite poset: its sup-irreducible elements form a poset P whose lower sets reconstruct the lattice, and every finite poset arises this way from its own lattice of lower sets.*

Let P be the finite poset of sup-irreducible elements of a finite frame. Then, by Birkhoff duality (`OrderIso.lowerSetSupIrred`), every finite frame is the lattice of lower sets of P . We fix the following notation, used throughout the rest of the paper. A lower set $L \subseteq P$ satisfies $p \in L, q \leq p \Rightarrow q \in L$, and an upper set is defined dually. Lower sets are always denoted L , and upper sets U (opens of a topological space are still U, V, W , as in traditional usage). The principal lower set at p is $\downarrow p = \{q : q \leq p\}$, the strict principal lower set at p is $\Downarrow p = \{q : q < p\}$, and the principal upper set at p is $\uparrow p = \{q : q \geq p\}$. We prove the finite representation theorem below on the lower sets of P . Transporting it across the isomorphism above then gives the theorem for every finite frame, not just lower sets^{Lean}. Let us define the mass of a point as the discrete derivative of v^{Lean} :

$$\text{mass}(p) = v(\downarrow p) - v(\Downarrow p).$$

The masses recover v and sum to 1:

Theorem 16 (finite representation^{Lean}). *For a finite P , any valuation v on $\text{Low}(P)$, and any $L \in \text{Low}(P)$, $v(L) = \sum_{p \in L} \text{mass}(p)$, and $\sum_p \text{mass}(p) = 1$.*

The proof works by induction on $|L|$: repeatedly remove a maximal point p from L , and use modularity applied to $(L \setminus \uparrow p, \downarrow p)$ to show each removal changes v by exactly $\text{mass}(p)$. The mass function is a classical probability mass function on P^{Lean} . And since lower sets are exactly the opens of the Alexandrov topology, where interior is the identity, the theorem also reads $v = \mu \circ \text{int}$, i.e. every valuation on a finite frame is the classical measure μ applied after taking the interior. In the finite case, this is the converse of the measure-theoretic bridge of Section 7. The intuitionistic mass also agrees with the classical notion of the mass of a point: reading $v = \mu \circ \text{int}$,

$$\text{mass}(p) = \mu(\downarrow p) - \mu(\Downarrow p) = \mu(\{p\}),$$

since $\downarrow p$ is the disjoint union of $\Downarrow p$ and $\{p\}$ and μ is defined on all subsets of P .

Theorem 16 represented v as a sum of point masses. The dual view represents v as a mixture of sharp point-valuations. For each $p \in P$, let δ_p be the valuation with $\delta_p(L) = 1$ if $p \in L$, and $\delta_p(L) = 0$ otherwise:

Theorem 17 (mixture characterization^{Lean}). *On a finite frame, every valuation v satisfies $v = \sum_p \text{mass}(p) \cdot \delta_p$.*

Together with the fact that any mixture of valuations is again a valuation^{Lean}, this theorem characterizes the valuations on a finite frame exactly: they are the finite convex mixtures of point-valuations, closed under mixing because modularity is linear. Section 6 shows modularity cannot instead be *derived* from more primitive axioms. It must be posited outright, and doing so is exactly what makes every valuation a mixture of sharp, all-or-nothing certainties.

4.2 Infinite Boundary Representation via Scott Continuity

Both halves of the finite theorem fail for infinite frames:

Example 18 (obstruction^{Lean}). On $\text{Low}(\mathbb{N})$, the indicator v of $\top$ is a valuation with $\text{mass}(p) = 0$ for every point p , and $v(\top) = 1$.

No point carries any of the mass. The failure is one of approximation: $v(\top)$ is not the supremum of $v(\downarrow n)$ over the finite stages. The $\top$ -indicator of Example 18 is not Scott-continuous^{Lean}, and Scott continuity is exactly what restores the representation:

Example 19 (recovery^{Lean}). Let v be a valuation on $\text{Low}(\mathbb{N})$. If $v(\top) = \sup_n v(\downarrow n)$ then $v(\top) = \sum_n \text{mass}(n)$.

In general, on any poset all of whose principal lower sets are finite, Scott continuity implies zero diffuse part without additional preconditions. However, subtraction in $[0, \infty]$ is truncated at 0, so if $\sum_p \text{mass}(p)$ were greater than $v(\top)$, the expression $v(\top) - \sum_p \text{mass}(p)$ would still evaluate to 0. Theorem 20 rules out $\sum_p \text{mass}(p) > v(\top)$ unconditionally, so the subtraction below is never truncated:

Theorem 20 (general decomposition^{Lean}). *For every valuation v on $\text{Low}(P)$ and an arbitrary poset P , $\sum_p \text{mass}(p) \leq v(\top)$.*

By Theorem 20 the remainder $v(\top) - \sum_p \text{mass}(p)$ is a genuine difference. Every valuation thus splits unconditionally into an *atomic* part carried by points ($\sum_p \text{mass}(p)$, which is always a sub-probability), and a *diffuse* (nonnegative) remainder $v(\top) - \sum_p \text{mass}(p)$. This is a constructive Lebesgue-style decomposition. The finite and Scott cases are the purely atomic extremes, and the $\top$ -indicator is the purely diffuse one. The proof of Theorem 20 re-runs the maximal-point induction on each finite subset of P , bounding its partial sum by $v(\top)$, then takes the supremum over all finite subsets to bound $\sum_p \text{mass}(p)$ itself. Its proof does not directly invoke an excluded-middle case split. However, it relies on the (classical) decidability of the ordering of the reals (a default in mathlib). Representing the diffuse part on non-spatial frames remains the open frontier. The obstruction there is again that the frame has *too few points*, the same one as behind the undecidability of Section 7.

5 The Dempster–Shafer Reading

Sections 2–4 independently arrive at the same structure as Dempster–Shafer (DS) theory [Dempster 1967; Shafer 1976], including sub-additive complements, a belief/plausibility interval, and mass functions. This section makes the identification exact in both statics and dynamics. In each case, it locates the precise point where excluded middle makes a difference.

5.1 Total Monotonicity via Constructive Inclusion–Exclusion

The $\neg\neg$ -stable (*regular*) elements of a frame form a Boolean algebra with meet $\sqcap$ and join $\neg\neg(a \sqcup b)$ (Glivenko). Let us call the resulting algebra the *Booleanization* of this set.

Definition 21 (belief function, plausibility function^{Lean}). Let B be a Boolean algebra. A DS *belief function* on B is a monotone map $\text{Bel} : B \rightarrow [0, 1]$ with $\text{Bel}(\perp) = 0$ and $\text{Bel}(\top) = 1$ that is *n-monotone* for every $n \geq 2$: for all $A_1, \dots, A_n \in B$,

$$\text{Bel}\left(\bigvee_{i=1}^n A_i\right) \geq \sum_{\emptyset \neq T \subseteq \{1, \dots, n\}} (-1)^{|T|+1} \text{Bel}\left(\bigwedge_{i \in T} A_i\right).$$

Its *plausibility function* is the map $\text{Pl} : B \rightarrow [0, 1]$ given by $\text{Pl}(A) = 1 - \text{Bel}(\neg A)$.

We show v , restricted to its Booleanization, is such a belief function. The 2-monotone case is immediate from modularity^{Lean}. The full tower follows from the following result:

Theorem 22 (inclusion–exclusion^{Lean}). *For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$, inclusion–exclusion holds with equality for the frame join:*

$$v\left(\bigsqcup_{i \in s} a_i\right) = \sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right).$$

We state this in Lean additively, with the odd-cardinality half of the signed sum set against the even-cardinality half, so that no $[0, \infty]$ subtraction occurs. The proof is by induction with modularity and frame distributivity.

Replacing the frame join $\bigsqcup_i a_i$ on the equality's other side with its double negation $\neg\neg(\bigsqcup_i a_i)$, the Booleanization's own join, yields the following inequality:

Theorem 23 (total monotonicity^{Lean}). *For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$:*

$$\sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right) \leq v\left(\neg\neg\left(\bigsqcup_{i \in s} a_i\right)\right).$$

The Booleanization's join is given by $\neg\neg \circ \sqcup$, and its meets of regular elements are again regular. Interpreted in this way, Theorem 23 states exactly the ∞ -monotone tower: v restricted to its Booleanization is a belief function.

The proof of Theorem 23 from Theorem 22 follows from the monotonicity along $a \leq \neg\neg a$. Here, inclusion–exclusion

does *not* fail constructively. It is exact for the intuitionistic disjunction. Applying the double-negation nucleus (a closure operator preserving finite meets) $\neg\neg$ to the join on the equality's right-hand side can only enlarge it. This follows from the fact that $a \leq \neg\neg a$ and that v is monotone, which turns the equality into the inequality of Theorem 23. The amount by which the right-hand side is enlarged is exactly the double-negation gap of Section 2.

Definition 21 is the classical notion, stated on an abstract Boolean algebra. In our setting each of its ingredients is already present: the Boolean algebra is the Booleanization, the belief function is v itself, restricted to the regular elements and evaluated against the Boolean join $\neg\neg \circ \sqcup$, and the plausibility function is $\text{Pl}(a) = 1 - v(\neg a)$. Additive probability does not become a belief function by weakening an axiom, since modularity for v is never relaxed. What changes is only the join used to state the tower. The belief–plausibility interval is governed by the slack:

Theorem 24 (interval width^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{Pl}(a) - v(a) = \text{slack}(a)$.*

5.2 The Conditioning Characterization

DS theory carries two update rules that coincide classically. *Geometric* conditioning renormalizes belief, and under the DS dictionary it is the localic posterior $v(a \sqcap b)/v(b)$ of Section 2.

Definition 25 (Dempster's rule of conditioning^{Lean}). For a valuation v on a frame Ω , elements $a, b \in \Omega$, and evidence of nonzero plausibility, $v(b) \neq 0$, *Dempster's rule of conditioning* [Shafer 1976] states that

$$v(a \parallel b) = \frac{v(a \sqcup \neg b) - v(\neg b)}{1 - v(\neg b)}.$$

The hypothesis $v(b) \neq 0$ makes the denominator nonzero, since $v(b) + v(\neg b) \leq 1$ gives $v(b) \leq 1 - v(\neg b)$.

That is, the DS conditioning rule discards the mass incompatible with the evidence and renormalizes by its plausibility. Both the geometric posterior of Section 2 and Dempster's rule of Definition 25 are again valuations. Their modularity is the frame distributivity $(a \sqcup \neg b) \sqcap (a' \sqcup \neg b) = (a \sqcap a') \sqcup \neg b$. Their disagreement is given by the *excluded-middle gap* $\text{emGap}(a, b) = v(a) - v(a \sqcap (b \sqcup \neg b))$, i.e. the mass of a outside b 's instance of excluded middle:

Lemma 26 (the join–complement identity^{Lean}). *For every valuation v on every frame Ω and every $a, b \in \Omega$, $v(a \sqcup \neg b) = v(\neg b) + v(a \sqcap b) + \text{emGap}(a, b)$.*

Dempster's numerator therefore exceeds Bayes's by exactly $\text{emGap}(a, b)$. This gap depends on a , but whether it vanishes at every a depends only on the slack at the evidence b :

Theorem 27 (the conditioning characterization^{Lean}). *For every valuation v on every frame Ω and every $b \in \Omega$ with*

$v(b) \neq 0$, $(v(a \parallel b) = v(a \sqcap b)/v(b))$ for every $a \in \Omega$ iff $\text{slack}(b) = 0$.

The hypothesis $\text{slack}(b) = 0$ eliminates the gap $\text{emGap}(a, b)$ for every a and makes the two normalizers coincide, since $1 - v(\neg b) = v(b)$. Conversely, testing at $a := b$, geometric conditioning gives certainty 1, while Dempster gives $v(b)/(1 - v(\neg b))$, which is 1 only if the slack vanishes. Theorem 27 is the valuation-relative analogue of Theorem 10: both are iff statements about the same proposition, $b \sqcup \neg b = \top$.

We call the frame-level fact *static* and the valuation-level fact *dynamic*: the former is an equation in the frame itself, between $b \sqcup \neg b$ and $\top$, with no valuation involved, while the latter tests the same proposition through v , since $\text{slack}(b) = 0$ is exactly $v(b \sqcup \neg b) = v(\top)$. Dempster and Bayes therefore coincide at b whenever v cannot distinguish $b \sqcup \neg b$ from $\top$, even if the two differ in the frame itself. In the measure model, Dempster and Bayes disagree precisely when the evidence's boundary carries mass.

6 Cox's Theorem

Cox's theorem [Cox 1946; Jaynes 2003] derives the probability calculus from conditions on plausibility. After Halpern's counterexample [Halpern 1999], its repaired forms [Arnborg and Sjödin 2000; Horn 2003] fix the calculus up to monotone regreduation. Every known derivation is classical, and Section 3 showed why: the negation axiom R3 is excluded middle. Dropping R3 leaves the product rule unaffected, since it is logic-free, and requires the sum rule to become an axiom rather than a theorem. Separately, the naive constructive statement, demanding a regreduation strictly monotone on all of $\mathbb{R}$, is unsatisfiable (Section 6.3).

6.1 A Cox Structure without Negation

Definition 28 (Cox model^{Lean}). A *Cox model* on a frame Ω is a plausibility assignment $\text{pl} : \Omega \times \Omega \rightarrow \mathbb{R}$, written $\text{pl}(a, b)$ for the plausibility of a given b , together with a *conjunction functional* $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, such that $\text{pl}(\cdot, c)$ is monotone for every c , the boundary conditions $\text{pl}(\perp, c) = 0$ and $\text{pl}(\top, c) = 1$ hold for every c , the product rule $\text{pl}(a \sqcap b, c) = F(\text{pl}(a, b \sqcap c), \text{pl}(b, c))$ holds, F is associative, and $x \mapsto F(x, y)$ is strictly monotone for every $y > 0$.

The missing R3 axiom can be observed in the omission of any constraint on the relation between $\text{pl}(\neg a, b)$ and $\text{pl}(a, b)$. The definition restricts the monotonicity of F to $0 < y$ to avoid a contradiction with the fact that the boundary axioms force $F(x, 0) = 0$ for all x . An explicit model on the non-classical chain $[0, \infty]$ satisfies this constraint^{Lean}, as $a = \top$ is provably decidable.

6.2 The Sum Rule Is Irreducible

Classically, additivity is *derived*: R3 turns disjunction into negated conjunction via De Morgan, and the product rule

does the rest. Constructively that route is closed, since $a \sqcup b \neq \neg(\neg a \sqcap \neg b)$ in general:

Example 29 (the sum rule is irreducible^{Lean}). Let Ω be the five-element frame of lower sets of the poset $o < a, o < b$ with a, b incomparable. Then there is a monotone, normalized assignment q on Ω that is additive on disjoint joins yet not modular, and no binary operation S on values satisfies $q(x \sqcup y) = S(q(x), q(y))$ for all x, y . Moreover, the pairs $(\downarrow a, \downarrow a)$ and $(\downarrow a, \downarrow b)$ have identical value-pairs $(\frac{1}{2}, \frac{1}{2})$ but joins valued $\frac{1}{2}$ and 1.

Unlike for a conjunction, a disjunction's plausibility cannot generally be expressed as its disjuncts' plausibilities via conditioning. So, a constructive Cox theorem *must* posit the sum-rule half. Modularity, which says that the calculus is closed under mixing certainties, is what posits this, justified by Theorem 17. The failure is not repaired by strengthening the logic short of Boolean: the counterexample frame satisfies weak excluded middle^{Lean}, since every nonempty lower set contains the bottom point o , so the irreducibility persists under that intermediate assumption.

6.3 Cox Model Theorem

A regratuation is a map $g : \mathbb{R} \rightarrow [0, \infty]$ with $g(0) = 0$. No such map is strictly monotone on all of $\mathbb{R}$, since strictness would force $g(-1) < g(0) = 0$, impossible in $[0, \infty]$. The repair (strictness only on $[0, 1]$, where plausibilities live) is exactly the same domain restriction that Halpern's critique of the classical theorem turns on. The corrected statement is:

Theorem 30 (the constructive Cox theorem^{Lean}). *For every frame Ω , every Cox model on Ω whose unconditional plausibility is modular regratuates to a valuation: there exist a valuation v on Ω and a strictly monotone function g on $[0, 1]$ with $g(0) = 0, g(1) = 1$, and for all $a \in \Omega, v(a) = g(\text{pl}(a, \top))$.*

The regratuation for the sum-rule half is straightforward ($g = \text{ENNReal.ofReal}$). The product-rule half, showing F regratuates to multiplication, rests on Aczél's associativity theorem [Aczél 1966], a statement of pure analysis whose type never mentions Ω . `mathlib` has no Aczél theorem, no Hölder-style embedding for ordered semigroups on intervals, and no Cauchy functional equation under monotonicity, so we build the analytic core from scratch, obtaining a normalized, strictly monotone multiplicative generator on the positive cone^{Lean}. The construction also weakens Hölder's classical hypotheses [Hölder 1901]: the Archimedean property is *derived* from continuity and positivity rather than assumed, and the interval form of divisibility follows from the intermediate value theorem^{Lean} applied to a plain divisibility hypothesis^{Lean}. The constructive/classical divide is therefore *entirely* in the sum rule: nothing in the product-rule apparatus mentions Ω . Appendix C gives the construction. The residual analytic gap is discussed in Section 14.

7 Two Grounding Models

Two concrete models demonstrate that our abstract theory is non-degenerate. One shows every classical measure *contains* an intuitionistic probability. The other shows the intuitionistic theory cannot collapse back without solving the halting problem.

7.1 Measures through the Interior Operator

Gödel–McKinsey–Tarski interpret intuitionistic logic in modal S4: propositions are *open* sets, and the modal necessity operator $\Box$ of S4 is interpreted as the topological interior, written $\text{int}(\cdot)$ below [Gödel 1933; McKinsey and Tarski 1944]. Our valuations quantify that same translation:

Theorem 31 (measures as valuations^{Lean}). *Let μ be a probability measure on a space $X, A \subseteq X$, and define $v(A) = \mu(\text{int } A)$. Since $\text{int } U = U$ for open U , v restricts on the frame $O(X)$ to $U \mapsto \mu(U)$, and this restriction is a valuation.*

Modularity is $\mu(U \cup V) + \mu(U \cap V) = \mu(U) + \mu(V)$. The slack of an open U is then $1 - \mu(U) - \mu(\text{int } U^c)$, which is the measure of the boundary ∂U . We can now prove that, in general, slack is exactly the boundary mass:

Theorem 32 (slack is boundary mass^{Lean}). *For any probability measure μ on a topological space X and any open U , $\text{slack}(U) = \mu(\partial U)$.*

The Heyting complement of U in $O(X)$ is the interior of its set complement. So, the undecided region, the complement of $U \sqcup \neg U$, is exactly $\overline{U} \setminus U = \partial U$. The general sub-additivity inequality of Section 2 still holds here^{Lean}. This identification of slack with boundary mass also exhibits the modal identification $\text{Bel} = P(\Box \cdot)$ of the DS literature [Ruspini 1986] arising from the logic itself. Theorem 16 is the converse in the finite case (in the general case, the converse is the open problem, see Section 4).

The same decomposition locates a failure of classical probability calculus in conditioning, that is, in updating a belief about an open A on evidence given by an open B . The Boolean complement of A splits into its Heyting complement and its boundary, $A^c = \text{int } A^c \sqcup \partial A$, so a classical posterior that treats A as confirmed-or-refuted overstates the refutation by the boundary mass the evidence retains:

Theorem 33 (classical overconfidence in refutation^{Lean}). *Let μ be a probability measure on a topological space X , let v be its restriction to $O(X)$, and let A, B be opens with $\mu(B) \neq 0$. Then*

$$\frac{\mu(A \cap B)}{\mu(B)} = v(A \mid B) \quad \text{and} \quad \frac{\mu(A^c \cap B)}{\mu(B)} = v(\neg A \mid B) + \frac{\mu(\partial A \cap B)}{\mu(B)}.$$

The excess term is the conditional slack of A given B , the mass that the evidence leaves neither confirmed nor refuted. The classical posterior does not discard this mass. It credits all of it to A^c , an event no finite observation confirms. For the halting event of Section 7.3 with $B = \top$, the classical

credence in non-termination is $1 - p$ while the constructive one is 0, and the difference $1 - p$ is exactly the slack.

To characterize weak excluded middle spaces, we give the (standard) definitions required:

Definition 34 (extremally disconnected). A topological space X is *extremally disconnected* if the closure of every open set is again open.

Definition 35 (Stone space, Stonean space). The *Stone space* of a Boolean algebra B is the space of ultrafilters on B , topologized by the clopen basis $\{U : b \in U\}$ for $b \in B$. *Stone duality* [Johnstone 1982] identifies Boolean algebras with exactly the compact, Hausdorff, totally disconnected spaces arising this way. A *Stonean space* is a compact Hausdorff space that is also extremally disconnected. Equivalently, it is the Stone space of a *complete* Boolean algebra.

Remark 36 (weak excluded middle on opens^{Lean}). Section 2 noted that imposing weak excluded middle on the whole frame collapses slack to pure double-negation instability. For $\Omega = O(X)$, this global condition holds exactly when X is extremally disconnected.

An immediate corollary gives the characterization:

Corollary 37 (Stonean characterization^{Lean}). *Given a compact Hausdorff space X , weak excluded middle on $O(X)$ holds exactly when X is a Stonean space.*

7.2 Slack Does Not Determine the Valuation

Submodularity (Theorem 7) is a necessary condition on slack. It is not sufficient: distinct valuations, each satisfying every axiom a valuation must satisfy, can share a slack function.

Example 38 (boundaries in the three-point space^{Lean}). Let $\text{ThreePt} = \{p_A, p_B, p_Z\}$ carry the topology generated by the two singletons $\{p_A\}$ and $\{p_B\}$, so its opens are exactly $\emptyset$, $\{p_A\}$, $\{p_B\}$, $\{p_A, p_B\}$, and the whole space. For every open U , $\partial U \subseteq \{p_Z\}$.

This is a fact about the topology alone, independent of any measure: p_Z has no proper open neighbourhood, so it is the only point any open's frontier can contain.

Example 39 (non-recovery^{Lean}). There exist valuations $v \neq v'$ on $O(\text{ThreePt})$ whose slack functions are identical: $\text{slack}_v = \text{slack}_{v'}$ as functions.

For weights $a, b, c \geq 0$ with $a + b + c = 1$, the point measure $\mu_{a,b,c} = a \delta_{p_A} + b \delta_{p_B} + c \delta_{p_Z}$, read through Section 7.1's bridge, gives a valuation $v_{a,b,c}$ on $O(\text{ThreePt})$. By Example 38 and Theorem 32, its slack at an open U is $\mu_{a,b,c}(\partial U)$, equal to c when $p_Z \in \partial U$ and to 0 otherwise: a quantity depending only on c , not on how the remaining mass splits between a and b . So $v_{1/4,1/4,1/2}$ and $v_{1/8,3/8,1/2}$ have identical slack functions while disagreeing at $\{p_A\}$, where they take the values $1/4$ and $1/8$. Both are fully modular, monotone, and normalized: nothing in Theorem 7 or its proof distinguishes them.

Weak excluded middle does not rescue recoverability. If anything, the fully classical case is the worst one for it. When X is extremally disconnected, the De Morgan gap vanishes (Section 7.1) and slack degenerates to the double-negation gap alone, but on a space where excluded middle holds outright that gap vanishes too, so slack is identically 0 for every valuation. The zero slack function is then shared by every probability measure on X , which is the most extreme failure of recovery available.

7.3 The Computability Guard

The theory developed here is non-degenerate only if there are spaces whose slack does not vanish identically. Computability supplies them:

Definition 40 (Sierpiński topology, semidecidable proposition). The *Sierpiński space* $\mathbb{S}$ is the type Prop of propositions carrying the topology with exactly three opens: $\emptyset$, the set $\text{halts} = \{q \mid q\}$ of true propositions, and $\mathbb{S}$ itself [Escardó 2004; Rosolini 1986]. A proposition is *semidecidable* if it is an open in this topology.

The topology encodes verifiability. A point of $\mathbb{S}$ is a proposition, say “this computation halts.” The one nontrivial observation available is membership in halts , established by producing a proof (running the computation to termination). No open captures the complementary observation, and on the frame $O(\mathbb{S}) = \{\perp, \text{halts}, \top\}$ that Σ_1 asymmetry is a theorem:

Theorem 41 (the Σ_1 asymmetry^{Lean}). $\neg \text{halts} = \perp$.

Every nonempty open of $\mathbb{S}$ contains True, so $\perp$ is the only open disjoint from halts . Thus, no nonempty open of $\mathbb{S}$ is evidence that the computation does not halt.

We now instantiate the measure-to-valuation construction of Theorem 31 at $X = \mathbb{S}$, allowing us to compute the slack:

Example 42 (halting slack^{Lean}). Let $\mu = p \cdot \delta_{\text{True}} + (1-p) \cdot \delta_{\text{False}}$, and let v be the resulting valuation. Then

$$\text{slack}(\text{halts}) = 1 - p = \mu(\partial \{q \mid q\}) = \mu\{\text{False}\}.$$

Since halts is already open, $v(\text{halts}) = \mu(\text{halts}) = p$. The DS “ignorance” of Section 5 is here the mass of the single point False, the one outcome no finite observation can ever resolve. This probability is a fact about the model, not about anyone's ignorance.

Chaitin's Ω . The same bridge scales to algorithmic information theory. Fix a universal prefix-free machine U , *prefix-free* in that no halting program of U is a proper prefix of another, and *universal* in that it simulates every other prefix-free machine. Give 2^ω , the space of infinite bit strings, the fair-coin measure μ . The event H of having a halting program of U as a prefix is open, since halting is established by a finite prefix (Definition 40), and prefix-freeness makes H a disjoint union of cylinders, so $v(H) = \mu(H) = \sum_{p \text{ halts}} 2^{-|p|}$.

exactly Chaitin's halting probability Ω [Chaitin 1975]. Its slack $\mu(\partial H)$ is the mass of strings approximated arbitrarily well by halting strings while never having a halting program as a prefix, and Ω 's classical uncomputability is the aggregate form of the pointwise obstruction above. The measure-theoretic core is mechanized on Cantor space with mathlib's infinite product measure: basic cylinders are clopen with measure $2^{-|w|_{\text{Lean}}}$, and a *finite* prefix-free set of words has pairwise disjoint cylinders of total measure $\sum_w 2^{-|w|_{\text{Lean}}}$, the Kraft-inequality content underlying Ω . The universal machine itself, and the countable extension of the finite Kraft sum, are computability theory beyond our scope.

Definition 43 (halting problem). For a code c and input n , the *halting problem* is the predicate $H(c, n) =$ “the machine with code c halts on input n ,” (ComputablePred).

Slack cannot be eliminated by simply declaring a classical answer for each code. The family of valuations defined below attempts to do this:

Definition 44 (sharp classical readout^{Lean}). For a given n , the *sharp classical readout* is the family of $\{0, 1\}$ -valued valuations $\rho_n(c) \in \text{Valuation}(O(\mathbb{S}))$, with $\rho_n(c)(\text{halts}) = 1$ if $H(c, n)$ holds, and $\rho_n(c)(\text{halts}) = 0$ otherwise.

The readout is sharp but not slack-free. Since $\neg \text{halts} = \perp$ (Theorem 41), a code that does not halt receives slack $1 - 0 = 1$, the maximum. Sharp here means that the classical answer is known, not that the complement rule holds on $O(\mathbb{S})$.

However, this family is provably not computable:

Theorem 45 (the sharp readout is uncomputable^{Lean}). For every n , the predicate $c \mapsto (\rho_n(c)(\text{halts}) = 1)$ is not computable.

The proof identifies the predicate with $H(\cdot, n)$ and applies the undecidability of the halting problem (ComputablePred.halting_problem).

No route back to a classical, decidable treatment survives contact with computability. Excluded middle guarantees that $H(c, n)$ is true or false for any fixed c , but that guarantee is free: it names no algorithm and exhibits no witness for which disjunct holds. Demanding an actual witness, uniformly in c , is exactly ρ_n 's sharp readout, and Theorem 45 shows no algorithm computes it. Aggregating over the whole family of halting programs, rather than reading off one instance, does not help either: Ω is a perfectly well-defined classical probability, yet Chaitin showed it uncomputable by an independent argument turning on algorithmic randomness [Chaitin 1975]. Neither the pointwise nor the aggregate route gives classical probability a way to compute a solution to the halting problem, and each is blocked by its own uncomputability result.

We can conclude that slack is not a defect of the constructive theory. On $O(\mathbb{S})$, forcing the complement rule at *halts* leaves

the single value $v(\text{halts}) = 1$, so every code is assigned certain termination. That assignment is computable but false of every non-halting code. Any assignment that avoids this error, giving $v(\text{halts}) < 1$ to each non-halting code, has positive slack at every such code, and the one that is moreover sharp is the uncomputable readout of Theorem 45. For the sum rule, this guard plays the role the explicit Cox-model instance plays for the product rule (Section 6): each axiom set in the paper comes with non-trivial models satisfying it.

8 Product Valuations and Kernel Composition

The structure through which classical probability supports sequential statistical reasoning is the Girly monad:

Definition 46 (Giry monad [Giry 1982]). The *Giry monad* G sends a measurable space X to the space $G(X)$ of probability measures on X . Its unit sends a point $x \in X$ to the point mass at x , and its multiplication sends a measure M on $G(X)$ to the measure $A \mapsto \int m(A) dM(m)$.

Probabilistic reasoning then composes in stages, and, when the monad is commutative, independent stages compose in either order. The question is whether the intuitionistic calculus supports the same structure.

A *frame homomorphism* is a map preserving finite meets and arbitrary joins. Pushforward along a frame homomorphism $f : \Omega \rightarrow \Omega'$ sends a valuation v on Ω' to $a \mapsto v(f(a))$ on Ω , and this preserves the valuation axioms, making $\text{Val}(-)$ a functor on locales, the localic counterpart of the functor part of the Girly monad [Giry 1982]. A frame homomorphism is the opposite of a locale map, so the functor laws are checked against identity and composition of frame homomorphisms^{Lean}. The point valuations δ_p (Section 4) provide the candidate (monad) unit. Theorem 16 shows that, on finite frames, Val agrees (on objects) with the classical finite-distribution monad on points. Flattening a valuation of valuations is integration, and integration in two variables requires product valuations. A *kernel* is a map assigning a valuation to each point of a locale, giving a conditional distribution. By Kock's identification of commutative monads with Fubini theorems [Kock 2012], kernels compose independently of order exactly when Val is a commutative monad.

A valuation u on a product frame is a *product* of valuations v and w when $u(A \times B) = v(A) \cdot w(B)$ for all rectangles $A \times B$. Let $\text{Low}(\mathbb{N} \times \mathbb{N})$ be the frame of lower sets of the grid $\mathbb{N} \times \mathbb{N}$, ordered component-wise. On this frame, products are unique:

Theorem 47 (product uniqueness^{Lean}). Let v, w be valuations on $\text{Low}(\mathbb{N})$. Any two valuations on $\text{Low}(\mathbb{N} \times \mathbb{N})$ that are products of v and w are equal.

By the inclusion-exclusion equality, two valuations agreeing on a $\sqcap$ -closed family agree on all finite joins of its members^{Lean}. The rectangles are closed under $\sqcap$, and in $\text{Low}(\mathbb{N} \times \mathbb{N})$

$\mathbb{N}$) every lower set is a *finite* union of rectangles^{Lean}, so agreement on rectangles is agreement everywhere. Valuations carry no continuity axiom (Section 4), so the σ -additivity driving the classical uniqueness argument for product measures has no analogue here. The order structure of the frame does that work instead.

Theorem 48 (product non-uniqueness^{Lean}). *On the powerset frame $\text{Set}(\mathbb{N} \times \mathbb{N})$ there exist valuations v, w and two distinct products of v and w .*

An *ultrafilter* on $\mathbb{N}$ is a collection of subsets, closed under intersection and superset, containing exactly one of S and its complement for every $S \subseteq \mathbb{N}$. It is *free* when it contains no finite set. The map assigning 1 to the members of an ultrafilter and 0 to every other subset is a valuation^{Lean}. A rectangle $S \times T$ belongs to an ultrafilter on the product exactly when S and T belong to its images under the two coordinate projections^{Lean}. A free ultrafilter $\mathcal{U}$ on $\mathbb{N}$ yields two ultrafilters on $\mathbb{N} \times \mathbb{N}$. A set belongs to the former when, for $\mathcal{U}$ -most first coordinates, its slice contains $\mathcal{U}$ -most second coordinates. A set belongs to the latter under the same test with the coordinates in the opposite order.

The two ultrafilters' valuations agree on *every* rectangle yet differ on the upper triangle $\{(m, m') \mid m < m'\}$ ^{Lean}. Integrating the same marginal data in the two orders thus disagrees. Fubini fails for valuations, as it does for non-additive integration [Denneberg 1994; Ghirardato 1997]. By Kock's correspondence, no canonical commutative multiplication exists for unrestricted valuations. This half of the dichotomy is not specific to intuitionism, since powerset-frame valuations are exactly the classical finitely additive probability measures [Rao and Rao 1983]. Dropping continuity is what admits the non-uniqueness. A monad therefore requires restricting either the frames or the valuations.

A first step in the second direction is to restrict Val to the valuations given as countable mixtures. For these, we can demonstrate that the kernels compose independently of order, the property that Theorem 48 refutes for unrestricted valuations:

Proposition 49 (countable kernel composition^{Lean}). 1. *A countable mixture of valuations is a valuation^{Lean}.*
 2. *A mixture of mixtures flattens to a mixture with product weights, which is discrete Chapman–Kolmogorov^{Lean}.*
 3. *The one-point mixture is the identity^{Lean}.*
 4. *Iterated countable mixing is order-independent^{Lean}.*

Restricted to countable-mixture presentations, the data of a monad are all in place: the category of locales, Val restricted to countable mixtures as the functor, the point valuations as the unit, and countable mixing as the multiplication. Proposition 49 verifies exactly the laws this needs: item 1 keeps the multiplication inside the fragment, item 2 is its associativity, item 3 its unit law, and item 4 its commutativity. What is not yet settled is whether this descends to valuations themselves,

since a valuation can admit more than one countable-mixture presentation, and the diffuse part of Section 4 is not captured by any countable presentation at all.

9 Valuations Beyond Frames

Definition 1 fixes the event algebra to be a frame: a complete lattice in which finite meets distribute over arbitrary joins $\bigsqcup_{i \in I} a_i$, the operation Heyting implication $\Rightarrow$ (the right adjoint to $a \sqcap (-)$) is built from. Definition 2's four axioms, normalization at $\perp$ and $\top$, monotonicity, and modularity $v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b)$, use only the *binary* join $\sqcup$; none of the four mentions the *arbitrary* join $\bigsqcup_{i \in I}$. Completeness is used only where later sections evaluate a valuation on an infinite join, as in the Scott-continuity results of Section 4, or build $\Rightarrow$ and $\neg$ from it, as in Section 3's slack. Section 14 leaves open how much of the construction is tied to frames specifically, and whether some ambient category sends an event algebra to a probability calculus in general. This section settles both questions for one fragment of the construction, the sharp/point correspondence of Section 3: it isolates the exact algebraic hypothesis that correspondence needs and shows the hypothesis cannot be dropped. It does not recover $\Rightarrow$ or $\neg$ without completeness: the adjoint defining $\Rightarrow$ is built from an arbitrary join of witnesses, so it is unavailable once completeness goes, even if only finite joins are missing. The valuations studied below are negation-free by construction, not frame valuations with completeness quietly removed.

The functor without completeness. A *bounded lattice* is a lattice with a least element $\perp$ and a greatest element $\top$; a *bounded lattice homomorphism* preserves $\perp$, $\top$, $\sqcup$, and $\sqcap$. Definition 2's axioms make sense over any bounded lattice Ω , complete or not:

Definition 50 (lattice valuation^{Lean}). A *lattice valuation* on a bounded lattice Ω is a monotone function $v : \Omega \rightarrow [0, \infty]$ with $v(\perp) = 0$, $v(\top) = 1$, and, for all $a, b \in \Omega$, $v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b)$.

```
structure LValuation (Ω : Type*) [Lattice Ω] [BoundedOrder Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b = toFun (a ⊔ b) + toFun (a ⊓ b)
```

A lattice valuation on a frame is exactly a valuation in the sense of Definition 2; the two notions differ only in how much structure the ambient Ω carries.

Section 8 pulls a valuation back along a frame homomorphism and checks the assignment is functorial (identity, composition). That argument pattern-matches on $\sqcup$, $\sqcap$, $\perp$, $\top$ alone and never touches $\sqcup$:

Proposition 51 (pullback without completeness^{Lean}). *For bounded lattices Γ, Ω , a bounded lattice homomorphism $f : \Gamma \rightarrow \Omega$, and a lattice valuation v on Ω , $v \circ f$ is a lattice valuation on Γ . This assignment sends the identity homomorphism to the identity and composition of homomorphisms to composition of the induced maps.*

$\text{Val}(-)$ is therefore a functor on bounded lattices with bounded lattice homomorphisms, of which the frame-level functor of Section 8 is the restriction to the subcategory of frames and frame homomorphisms. Completeness contributes nothing to functoriality.

Separation needs only distributivity. Theorem 12 characterizes what a sharp valuation is, a complement-indicator of a prime ideal, but not whether there are enough of them to distinguish elements of Ω . That is a separate fact:

Theorem 52 (sharp valuations separate points^{Lean}). *For every frame Ω and $a, b \in \Omega$ with $a \not\leq b$, there is a sharp valuation v with $v(a) = 1$ and $v(b) = 0$.*

The proof runs the same prime-ideal separation theorem that built the counterexample of Theorem 10 (`DistribLattice.prime_ideal_of_disjoint_filter_ideal`), satisfy all three pairwise sum-to-one equations at once, and a two-element codomain cannot support that: fixing any one value propagates through the other two equations to a contradiction with the third. The collapse on M_3 is therefore not a failure of separation, points that happen to coincide, but a failure of existence: there is no all-or-nothing valuation on M_3 to separate anything with, and the first qualifying condition of Section 14 is unsatisfiable outright.

Tracing the proof of Theorem 52 shows it never invokes $\sqcup$; the only place frame structure enters is `DistribLattice.prime_ideal_of_disjoint_filter_ideal` itself, which needs distributivity and nothing beyond it:

Theorem 53 (separation over distributive lattices^{Lean}). *For every distributive bounded lattice Ω and $a, b \in \Omega$ with $a \not\leq b$, there is a sharp lattice valuation v on Ω with $v(a) = 1$ and $v(b) = 0$.*

Theorem 53 reproves Theorem 52 with the hypothesis `Order.Frame` weakened to `DistribLattice`, and nothing

else about the argument changes. The condition doing the work in the sharp/point correspondence is distributivity; completeness is not load-bearing for either half of it.

Distributivity is not a proof artifact. The natural question is whether Theorem 53's hypothesis is a genuine dividing line or an artifact of routing the argument through a mathlib lemma named for distributive lattices. It is genuine, and the failure it guards against is total, not merely a failure of separation:

Definition 54 (the diamond^{Lean}). M_3 is the five-element lattice with a bottom z , a top o , and three pairwise-incomparable atoms a, b, c , with $x \sqcup y = o$ and $x \sqcap y = z$ for every two distinct atoms x, y .

M_3 is the smallest lattice that is not distributive: $a \sqcap (b \sqcup c) = a \sqcap o = a$, while $(a \sqcap b) \sqcup (a \sqcap c) = z \sqcup z = z$. It is modular, so this failure is specific to distributivity rather than some coarser defect of the order.

Theorem 55 (no sharp valuation on the diamond^{Lean}). *No lattice valuation on M_3 is sharp.*

Modularity applied to each of the three pairs of distinct atoms forces $v(x) + v(y) = 1$, since $x \sqcup y = \top$ and $x \sqcap y = \perp$. If v were sharp, $v(a), v(b), v(c) \in \{0, 1\}$ would have to satisfy all three pairwise sum-to-one equations at once, and a two-element codomain cannot support that: fixing any one value propagates through the other two equations to a contradiction with the third. The collapse on M_3 is therefore not a failure of separation, points that happen to coincide, but a failure of existence: there is no all-or-nothing valuation on M_3 to separate anything with, and the first qualifying condition of Section 14 is unsatisfiable outright.

Where this leaves the boundary. Frames are complete distributive lattices. Theorem 53 shows completeness is not needed for the sharp/point correspondence's separation half, and Theorem 55 shows distributivity cannot be given up in exchange. The event algebras on which sharp valuations reconstruct the order, the order-theoretic core of the first qualifying condition of Section 14, are exactly the distributive bounded lattices, no more and no less. The second qualifying condition and the negation-dependent machinery of Sections 3–7, slack, the complement bound, the De Morgan gap, are untouched by this section and remain tied to frames: nothing here shows, or claims, that $\neg$ or $\Rightarrow$ survives dropping completeness. This bears on the orthomodular event lattices of quantum probability discussed in Section 13: giving up distributivity, as those lattices do, does not merely change which valuations exist, it can remove sharp valuations altogether, so the construction of this paper does not, without further modification, have anything to say about that setting even at the order-theoretic level examined here. Whether some

weaker notion of certainty survives on non-distributive lattices, in place of the $\{0, 1\}$ -valued sharp valuations examined here, is not addressed by this section.

A second boundary: grading a non-idempotent tensor.

Section 13 notes that linear logic's quantale tensor $\otimes$ is not the lattice meet $\sqcap$ this section's valuations are built from, so Definition 50's modularity axiom does not even parse for $\otimes$ without first deciding what to write in $\sqcap$'s place. Granting the substitution provisionally, to see what it costs, shows the obstruction runs deeper than a notational mismatch:

Definition 56 (tensor valuation^{Lean}). Fix a join-semilattice Ω and a binary operation $\otimes$ on Ω . A *tensor valuation* for $\otimes$ is a function $v : \Omega \rightarrow [0, \infty]$ such that, for all $a, b \in \Omega$, $v(a) + v(b) = v(a \sqcup b) + v(a \otimes b)$.

Theorem 57 (the transplant collapses^{Lean}). For every join-semilattice Ω , binary operation $\otimes$ on Ω , and tensor valuation v for $\otimes$, if $v(a) < \infty$ then $v(a \otimes a) = v(a)$.

Setting $b := a$ in the defining identity and using only $a \sqcup a = a$, join-idempotence, already gives $v(a) + v(a) = v(a) + v(a \otimes a)$, hence the claim by cancellation. No property of $\otimes$ is used: not associativity, not a unit, not distributivity over $\sqcup$. This is a sharper failure than Theorem 55's: M_3 shows meet-based modularity can fail to admit a sharp valuation on one uncooperative lattice, while Theorem 57 shows the $\otimes$ -based transplant admits no valuation separating a from $a \otimes a$ on any Ω and for any $\otimes$, idempotent or not. A non-idempotent $\otimes$, linear logic's whole point, is exactly what such a valuation could never distinguish from $\sqcap$'s behavior.

Example 58 (a concrete witness^{Lean}). Take $\Omega = \mathbb{N}$ with $\sqcup = \max$ and $\otimes = +$: addition is associative and commutative but not idempotent ($1 + 1 = 2 \neq 1$), so the hypothesis of Theorem 57 is not vacuous. Substituting $a = 0$, where the join's unit and the tensor's unit coincide, forces $v(n) = v(0)$ for every n : every tensor valuation for $+$ is constant.

This is not a toy chosen merely by analogy with linear logic. Every quantale, the event algebra Section 13 attaches to linear logic, is by definition a complete lattice equipped with an associative $\otimes$ that need not be idempotent, exactly the data $(\Omega, \sqcup, \otimes)$ Theorem 57 quantifies over, with the quantale's own join playing the role of $\sqcup$. Example 58 is a quantale itself ($+$ distributes over $\max$), kept only because it is small enough to check by hand: it fails idempotence for the same reason $\otimes$ does, no algebraic law identifies two copies of a resource with one.

Corollary 59 (grading and discrimination are exclusive^{Lean}). For every join-semilattice Ω , binary operation $\otimes$ on Ω , and $a \in \Omega$ with $v(a) < \infty$: no v is both a tensor valuation for $\otimes$ and satisfies $v(a \otimes a) \neq v(a)$.

This restates Theorem 57 as the dichotomy it really is. A candidate probabilistic semantics for a non-idempotent

$\otimes$ must choose between being graded by Definition 50's compositional law and discriminating the one thing a non-idempotent $\otimes$ exists to discriminate, a from $a \otimes a$; it cannot keep both. The probabilistic semantics that does exist for linear logic, probabilistic coherence spaces [Danos and Ehrhard 2011], keeps discrimination: two independent copies of a resource are not scored as interchangeable with one. Corollary 59 shows the price of that choice is forced, not chosen: probabilistic coherence spaces cannot also be graded by this paper's modularity law, on pain of losing exactly the discrimination they exist to keep, so they instead evaluate proofs with a classical, Kolmogorov-style structure over $\otimes$'s type-level behavior, never a valuation of propositions in Definition 50's sense. Nothing here is a missed opportunity on either side; the dichotomy forecloses the combination before any further axiom of Definition 50 is even reached. Section 10 replaces these axioms by a construction that applies to linear logic.

10 Probability Calculi from Certain States

Corollary 59 shows that linear logic admits no grading by Definition 50's axioms that distinguishes a from $a \otimes a$. The obstruction is in the modularity axiom, which uses the lattice operations of the event algebra. This section replaces the three axioms of Definition 2 by a construction that uses no lattice operations. It applies to any logic presented by a set of certain states. On finite frames, and in particular on finite Boolean algebras, it returns exactly the valuations of Definition 2. For every formula of multiplicative-additive linear logic over finite data types, its output is contained in the probabilistic coherence space (Definition 66) of the formula, and the certain elements of that space are cliques of Girard's coherence space. At implications between data types it returns the probabilistic coherence space. At the exponential $! \text{Bool}$ it returns a strict subset.

10.1 The Construction

Definition 60 (presentation by certain states, mixtures^{Lean}). A *presentation* is a set E of events together with a set S of $\{0, 1\}$ -valued functions $E \rightarrow [0, \infty]$, the *certain states*. Its *calculus* $\text{Mix}(S)$ is the set of finite convex combinations $\sum_i w_i s_i$ with $s_i \in S$, $w_i \in [0, \infty]$, and $\sum_i w_i = 1$.

For classical logic the certain states are the two-valued models, for intuitionistic logic on a frame they are the sharp valuations (Theorem 12), and for a type of linear logic they are the deterministic programs of that type (Section 10.2). Two properties hold for every presentation.

Theorem 61 (certain elements and mixing^{LeanLean}). For every presentation (E, S) : (i) $a \{0, 1\}$ -valued x lies in $\text{Mix}(S)$ iff $x \in S$, and (ii) $\text{Mix}(\text{Mix}(S)) = \text{Mix}(S)$.

Part (i) is the first condition of Section 1, that the $\{0, 1\}$ -valued elements of the calculus reconstruct the logic. In a

presentation the logic is given by S , so the condition says that mixing creates no new $\{0, 1\}$ -valued elements. The proof: if $x = \sum_i w_i s_i$ is $\{0, 1\}$ -valued and $w_j \neq 0$, then at each e with $x_e = 0$ every s_i with $w_i \neq 0$ has $s_i(e) = 0$, and at each e with $x_e = 1$ every such s_i has $s_i(e) = 1$, since $\sum_i w_i = 1$ and $s_i(e) \leq 1$. So $s_j = x$, and $x \in S$. Part (ii) states that a convex combination of elements of $\text{Mix}(S)$ is again in $\text{Mix}(S)$. For valuations on a frame this is the closure of valuations under mixing of Section 4^{Lean}. The second condition of Section 1, that the classical case returns the classical theory, is Theorem 64 below.

The three axioms of Definition 2 are instances of one form:

Definition 62 (affine law^{Lean}). For finite E , an *affine law* is given by $a, b : E \rightarrow [0, \infty]$ and $k, k' \in [0, \infty]$. A function $x : E \rightarrow [0, \infty]$ satisfies it if $\sum_e a_e x_e + k \leq \sum_e b_e x_e + k'$.

Monotonicity $x_a \leq x_b$, normalization ($x_\top \leq 1$ and $1 \leq x_\perp$), and modularity (two opposite inequalities) are affine laws.

Theorem 63 (completeness^{Lean}). For finite E and every set S of $\{0, 1\}$ -valued functions on E , $\text{Mix}(S)$ is the set of functions satisfying every affine law that every $s \in S$ satisfies.

The inclusion from left to right holds because affine laws are preserved by convex combinations^{Lean}. The converse separates a point outside the convex hull of S by a hyperplane, using mathlib's geometric Hahn–Banach theorem, and reads the hyperplane off as an affine law. Theorem 63 is the generalization of the three axioms: the axioms of the calculus of a presentation are the affine laws valid on its certain states, and nothing else. On frames these are the three axioms of Definition 2:

Theorem 64 (frames^{Lean}). For a finite poset P , the valuations on the frame $\text{Low}(P)$ of lower sets of P are exactly Mix of its sharp valuations.

The proof combines Theorem 17 with the fact that a sharp valuation on $\text{Low}(P)$ is a point valuation δ_p ^{Lean}. Every finite frame is of the form $\text{Low}(P)$ (Section 4), and finite Boolean algebras are the case of a discrete P , where the valuations are the probability mass functions on P . So the construction reduces to the valuation calculus on finite intuitionistic event algebras and to Kolmogorov's calculus on finite classical ones.

For infinite E the laws are taken with finite support. A *window* is a finite set $F \subseteq E$ of events, and a *finite law* is an affine law on some window, applied to the restrictions of functions to F .

Theorem 65 (barycenters^{LeanLean}). Let S be a set of $\{0, 1\}$ -valued functions on an arbitrary set E . For $x : E \rightarrow [0, \infty]$ the following are equivalent: (i) x satisfies every finite law valid on S ; (ii) for every window F , the restriction of x to F lies in Mix of the restrictions of S to F ; (iii) there is a Borel probability measure μ on $\{0, 1\}^E$, with the product topology, such that

$x_e = \mu\{\sigma \mid \sigma_e = 1\}$ for every $e \in E$, and $\mu(\text{bad}_F) = 0$ for every window F , where bad_F is the set of σ whose restriction to F is the restriction of no $s \in S$.

The equivalence of (i) and (ii) applies Theorem 63 to each window. For (ii) $\Rightarrow$ (iii), the mixture on a window F gives a probability measure μ_F on $\{0, 1\}^E$, a finite convex combination of point masses at certain states. The windows are directed by inclusion. The probability measures on the compact space $\{0, 1\}^E$ form a compact space in the weak topology², so the family $(\mu_F)_F$ has a cluster point μ . A *cylinder set* is a set of σ determined by the values of σ on a window. Cylinder sets are clopen, so $v \mapsto v(C)$ is continuous in the weak topology for every cylinder set C . The sets $\{\sigma \mid \sigma_e = 1\}$ and bad_F are cylinder sets, and μ_F has the required values on them once $e \in F$, so μ has them. For (iii) $\Rightarrow$ (i), a finite law valid on S holds at μ -almost every σ , and integrating it against μ gives the law at x .

$\text{Mix}(S)$ is the least convex set containing S . Criteria (1) and (2) of Section 1 do not single it out: the probabilistic coherence space !Bool satisfies (1) and strictly contains the calculus (Theorem 68). Whether a further criterion selects $\text{Mix}(S)$, and how the criteria should treat logics without certain states (Section 12), is open.

10.2 Linear Logic

Definition 66 (probabilistic coherence space [Danos and Ehrhard 2011]^{Lean}). For a set W and a set Q of functions $W \rightarrow [0, \infty]$, let $Q^\perp = \{y : W \rightarrow [0, \infty] \mid \sum_w x_w y_w \leq 1 \text{ for all } x \in Q\}$. A *probabilistic coherence space* (PCS) with *web* W is a set P of functions $W \rightarrow [0, \infty]$ such that $P = P^{\perp\perp}$, every $w \in W$ has some $x \in P$ with $x_w \neq 0$ (totality), and $\sup_{x \in P} x_w < \infty$ for every $w \in W$ (boundedness).

We use finite webs. For a finite set X , the *data type* X is the PCS of $x : X \rightarrow [0, \infty]$ with $\sum_a x_a \leq 1$. The PCS $X \multimap Y$ on $X \times Y$ is the set of t with $\sum_b \sum_a x_a t(a, b) \leq 1$ for every x in X , equivalently $\sum_b t(a, b) \leq 1$ for every a ^{Lean}. Both satisfy the three conditions^{Lean}. A *coherence space* [Girard 1987] is a set with a reflexive symmetric relation, its coherence, and a *clique* is a set of pairwise coherent elements. In the coherence space $X \multimap Y$ of two data types, two points (a, b) and (a', b') cohere iff they are equal or $a \neq a'$.

Probabilistic coherence spaces interpret every formula of linear logic, including the exponentials, and model probabilistic PCF [Danos and Ehrhard 2011]. The construction of Definition 60 applies to every such type, with the $\{0, 1\}$ -valued elements of its probabilistic coherence space as certain states. We compare the two at the types where we have results. At first order they agree. The exponential !Bool separates them already at first order. Section 10.3 proves the facts that hold

²In mathlib this is an instance proved from Prokhorov's theorem.

at every multiplicative–additive formula. Whether the construction agrees with the probabilistic coherence space at other types is open.

Theorem 67 (first order^{LeanLean}). $X \multimap Y = \text{Mix}\{\det f \mid f : X \rightarrow Y_\perp\}$, where $\det f(a, b) = 1$ if $f(a) = b$ and 0 otherwise, and $Y_\perp$ adds a divergent value. A $\{0, 1\}$ -valued t lies in $X \multimap Y$ iff its support is a clique of the coherence space $X \multimap Y$.

At first order the certain states are the deterministic partial programs, they are the cliques of the coherence-space model, and the calculus is the PCS.

The exponential separates the two at first order. The PCS $! \text{Bool}$ has web the finite multisets over $\{t, f\}$, and is the biorthogonal closure of the promotions $x^!(i, j) = x_t^i x_f^j$ of the elements x of Bool , where (i, j) is the multiset with i copies of t and j copies of f .

Theorem 68 (exponential^{LeanLean}). Every $\{0, 1\}$ -valued element of $! \text{Bool}$ vanishes at every (i, j) with $i, j \geq 1$. The promotion of $x = (\frac{1}{2}, \frac{1}{2})$ takes the value $\frac{1}{4}$ at $(1, 1)$, so it lies in $! \text{Bool}$ and not in Mix of the $\{0, 1\}$ -valued elements.

The first statement uses the dual element $4\delta_{(i,j)}$, which is admissible because $x_t + x_f \leq 1$ implies $4x_t x_f \leq 1$. The $\{0, 1\}$ -valued elements are therefore supported on the multisets whose underlying set is a clique of Bool . These underlying sets are the points of the web of $! \text{Bool}$ in Girard’s coherence spaces. Danos and Ehrhard [Danos and Ehrhard 2011] observe that elements of the PCS $! \text{Bool}$ are nonzero on multisets whose underlying set is not a clique. Theorem 68 shows that such elements are not mixtures of certain elements.

10.3 All Multiplicative–Additive Formulas

Definition 69 (formulas and their semantics^{Lean}). The formulas are generated from the data types $\mathbf{n}$, for $n \in \mathbb{N}$, by linear negation $A^\perp$, $A \otimes B$, and $A \& B$. Write $A \wp B = (A^\perp \otimes B^\perp)^\perp$, $A \oplus B = (A^\perp \& B^\perp)^\perp$, and $A \multimap B = (A \otimes B^\perp)^\perp$. The web $|A|$ is $\{1, \dots, n\}$ for $\mathbf{n}$, $|A|$ for $A^\perp$, $|A| \times |B|$ for $A \otimes B$, and $|A| \sqcup |B|$ for $A \& B$. The PCS $P(A)$ is the data type for $\mathbf{n}$, $P(A)^\perp$ for $A^\perp$, $\{x \otimes y \mid x \in P(A), y \in P(B)\}^{\perp\perp}$ for $A \otimes B$, where $(x \otimes y)(a, b) = x_a y_b$, and the set of z whose restrictions to $|A|$ and $|B|$ lie in $P(A)$ and $P(B)$ for $A \& B$ [Danos and Ehrhard 2011]. The coherence $\asymp_A$ is equality on $\mathbf{n}$, and $s \asymp_{A^\perp} t$ iff $s = t$ or not $s \asymp_A t$. On $A \otimes B$ it is componentwise. On $A \& B$ it is $\asymp_A$ and $\asymp_B$ on the two components, and any two points from different components cohere [Girard 1987].

Theorem 70 (every formula^{LeanLeanLeanLean}). For every formula A : (i) $P(A)$ is a PCS; (ii) the indicator of every point and of every pair of distinct coherent points lies in $P(A)$, and $x_s + x_t \leq 1$ for every $x \in P(A)$ and every pair of distinct incoherent points s, t ; (iii) every $\{0, 1\}$ -valued element of $P(A)$ is the indicator of a clique of the coherence space A ; (iv) Mix of the $\{0, 1\}$ -valued elements of $P(A)$ is contained in $P(A)$, and its $\{0, 1\}$ -valued elements are exactly those of $P(A)$.

Part (ii) is proved by simultaneous induction on A . Negation exchanges the statement about coherent pairs with the statement about incoherent pairs. For $A \otimes B$, the indicator of a coherent pair is bounded by the tensor of indicators of its projections, and a pair incoherent in $A \otimes B$ is incoherent in one component, where the bound of A or B applies. Part (iii) follows from (ii): two incoherent points in the support of a $\{0, 1\}$ -valued element would give it total 2 on a pair bounded by 1. Part (iv) is Theorem 61 together with the closure of an orthogonal under mixing^{Lean}.

The semantics of Definition 69 computes the spaces of Section 10.2: $P(\mathbf{n} \multimap \mathbf{m})$ is the PCS $X \multimap Y$ of that section for $X = \{1, \dots, n\}$ and $Y = \{1, \dots, m\}$ ^{Lean}. So Theorem 67 is a statement about these formulas^{Lean}.

The exponentials, whose webs are infinite, are not covered by Definition 69. Theorem 68 treats $! \text{Bool}$ separately.

11 Second-Order Linear Types

This section uses the construction of Section 10 at the second-order type of a program with two functional arguments, where it returns a strict subset of the probabilistic coherence space. By computation, not formalized, the construction and the probabilistic coherence space agree on $(\text{Bool} \multimap \text{Bool}) \multimap \text{Bool}$ and $((\text{Bool} \multimap \text{Bool}) \multimap \text{Bool}) \multimap \text{Bool}$, the two iterated implications into Bool we checked. The type T below is the smallest we found where they differ without exponentials.

11.1 Randomized Sequential Programs

At second order, let $T = ((X \multimap Y) \otimes (X' \multimap Y'))^\perp$, the type of a program that receives $f : X \multimap Y$ and $g : X' \multimap Y'$, uses each once, and halts or diverges. We take T to be the set of $t : (X \times Y) \times (X' \times Y') \rightarrow [0, \infty]$ with $\sum_{p,q} y_p t(p, q) z_q \leq 1$ for all y in $X \multimap Y$ and z in $X' \multimap Y'$, the orthogonal of the products $y \otimes z$. For $X = Y = X' = Y' = \text{Bool}$, Lean checks that T satisfies the three PCS conditions^{Lean}. A *randomized sequential program* queries one argument first. Querying f first, it chooses the input a with probability $\mu(a)$, after the answer b chooses the input c to g with probability $\kappa(a, b, c)$, and after the answer d halts with probability $\rho(a, b, c, d)$. Its value at $((a, b), (c, d))$ is $\mu(a)\kappa(a, b, c)\rho(a, b, c, d)$ plus the analogous term for programs that query g first, with $\sum \mu \leq 1$ over both orders, $\sum_c \kappa(a, b, c) \leq 1$, and $\rho \leq 1$. It is *deterministic* if its values are in $\{0, 1\}$.

Theorem 71 (second order^{LeanLean}). For nonempty finite X, X' and finite Y, Y' : (i) a $\{0, 1\}$ -valued z lies in T iff its support lies in a tree $\{((a, b), (c, d)) \mid b \in Y, d \in Y'\}$ for some $a \in X$ and $c : Y \rightarrow X'$, or in a tree $\{((a, d), b), (c, d))\}$ that queries g first, and these are exactly the deterministic sequential programs^{Lean}; (ii) Mix of the $\{0, 1\}$ -valued elements of T is exactly the set of randomized sequential programs.

Part (ii) contains both directions of Kuhn’s theorem for this type [Kuhn 1953]: every randomized sequential program

is a mixture of deterministic ones^{Lean}, and every mixture of randomized sequential programs is a randomized sequential program^{Lean}. The second direction conditions the choices at each node on the history, and nodes reached with probability 0 are given weight 0.

The calculus is strictly smaller than the PCS. Call two distinct web points of T *exclusive* if a single pair of deterministic partial programs (f, g) produces both, that is, if their first components cohere in $X \multimap Y$ and their second components cohere in $X' \multimap Y'$. The *exclusivity graph* of T has the web points as vertices and the exclusive pairs as edges.

Theorem 72 (strict inclusion^{LeanLean}). *Let X, Y, X', Y' have at least two elements each. There are five web points of T such that the function with value $\frac{1}{2}$ on them and 0 elsewhere lies in T and is not in Mix of the $\{0, 1\}$ -valued elements of T . For $X = Y = X' = Y' = \text{Bool}$, the total weight on the five points is at most 2 for every randomized sequential program, 2 is attained, and it is at most $\frac{5}{2}$ on T , attained by the function above.*

Over Bool , writing (a, b) for the observation $f(a) = b$, the five points are $((t, t), (t, t))$, $((t, f), (f, f))$, $((f, t), (t, f))$, $((f, t), (f, t))$, and $((f, f), (f, f))$. They induce a 5-cycle in the exclusivity graph. Lean checks the two facts the proof uses, that the induced subgraph has no triangle and no three pairwise non-adjacent vertices^{Lean}, and embeds the five points into larger types by injections from Bool .

The semantics of Definition 69 computes $T: P(((\mathbf{n} \multimap \mathbf{m}) \otimes (\mathbf{n}' \multimap \mathbf{m}'))^\perp)$ is T for $X = \{1, \dots, n\}$, $Y = \{1, \dots, m\}$, $X' = \{1, \dots, n'\}$, $Y' = \{1, \dots, m'\}$ ^{Lean}. So Theorem 71(ii) and the strict inclusion of Theorem 72 are statements about these formulas^{Lean}.

11.2 A Clique Outside the PCS

The converse of Theorem 70(iii) fails. Let $T = ((2 \multimap 2) \otimes (2 \multimap 2))^\perp$. On the five points of Theorem 72, two distinct points cohere in T iff they are not exclusive. Order them as $c_0, \dots, c_4$ along the 5-cycle of the exclusivity graph. The coherence graph on them is the complement of this cycle, again a 5-cycle. Two distinct points of $T \otimes T$ are incoherent iff they are incoherent in one component. A clique of $(T \otimes T)^\perp$ contained in $\{c_0, \dots, c_4\}^2$ is therefore a stable set of the strong product of the two coherence graphs, a copy of $C_5 \boxtimes C_5$. The set $\{(c_i, c_{2i}) \mid i \in \mathbb{Z}/5\}$ is such a stable set, of size 5. It is Shannon's stable set, which shows that the Shannon capacity of C_5 is at least $\sqrt{5}$ [Lovász 1979].

Theorem 73 (a clique outside the PCS^{LeanLean}). *Shannon's set is a clique of the coherence space $(T \otimes T)^\perp$, and its indicator is not an element of $P((T \otimes T)^\perp)$.*

The function z of Theorem 72, with value $\frac{1}{2}$ on the five points, lies in $P(T)$, so $z \otimes z$ lies in $P(T \otimes T)$. Its pairing with the indicator of Shannon's set is $5 \cdot \frac{1}{4} = \frac{5}{4} > 1$. Two choices of certain states for a formula are therefore different

from $(T \otimes T)^\perp$ on: the $\{0, 1\}$ -valued elements of the PCS, for which the calculus is contained in the PCS at every formula (Theorem 70(iv)), and the cliques of the coherence space, for which it is not.

11.3 When the Calculus Equals the PCS

For a finite graph G , a *clique* is a set of pairwise adjacent vertices and a *stable set* a set of pairwise non-adjacent vertices. Let $\text{STAB}(G)$ be Mix of the indicator functions of stable sets, and $\text{QSTAB}(G)$ the set of $x : V \rightarrow [0, \infty]$ with $\sum_{v \in C} x_v \leq 1$ for every clique C [Grötschel et al. 1988].

Theorem 74 (graph polytopes^{LeanLean}). *$T = \text{QSTAB}(G)$ and Mix of the $\{0, 1\}$ -valued elements of T is $\text{STAB}(G)$, for G the exclusivity graph of T .*

Theorem 75 (odd holes and antiholes^{LeanLean}). *If $\text{STAB}(G) = \text{QSTAB}(G)$, then G contains no induced cycle of odd length $2k + 1 \geq 5$ and no induced complement of one. In particular, if Mix of the $\{0, 1\}$ -valued elements of T equals T , the exclusivity graph of T has neither.*

A graph with neither is *Berge*. A graph is *perfect* if every induced subgraph has chromatic number equal to its largest clique size. The converse of Theorem 75 holds: $\text{STAB}(G) = \text{QSTAB}(G)$ iff G is perfect [Chvátal 1975; Lovász 1972], and G is perfect iff it is *Berge*, by the strong perfect graph theorem [Chudnovsky et al. 2006]. The converse is not formalized.

The *independence number* $\alpha(G)$ is the largest size of a stable set, and the *fractional packing number* $\alpha^*(G)$ is the largest value of $\sum_v x_v$ over $\text{QSTAB}(G)$. The bounds 2 and $\frac{5}{2}$ of Theorem 72 are $\alpha(C_5)$ and $\alpha^*(C_5)$. Cabello, Severini and Winter [Cabello et al. 2014] show that in a contextuality scenario with exclusivity graph G the noncontextual bound is $\alpha(G)$, the quantum bound is the Lovász number $\vartheta(G)$ [Lovász 1979], and the bound for theories satisfying only exclusivity is $\alpha^*(G)$. For C_5 this is the KCBS inequality [Klyachko et al. 2008], with $\vartheta(C_5) = \sqrt{5}$. A calculus for T corresponding to the theta body of the exclusivity graph [Grötschel et al. 1988], which lies between STAB and QSTAB , is not constructed here.

12 Quantum Event Logic

In quantum logic the events are the closed subspaces of a Hilbert space [Birkhoff and von Neumann 1936]. We restrict to rays, the one-dimensional subspaces, of $\mathbb{R}^d$. A *context* is an orthogonal basis, a maximal set of pairwise orthogonal rays, whose events can be observed jointly. A *quantum state* assigns to each ray a number in $[0, \infty]$ such that the numbers on every context sum to 1. For a unit vector ψ , the *Born rule* $x_v = (\psi \cdot v)^2 / (v \cdot v)$ gives such an assignment. For $d \geq 3$, Gleason's theorem [Gleason 1957] states that every quantum state on all rays comes from a density matrix, a mixture of Born-rule assignments.

The presentation of Definition 60 takes the rays as events and, as certain states, the $\{0, 1\}$ -valued quantum states, which give exactly one ray of every context the value 1. By the Kochen–Specker theorem [Kochen and Specker 1967], for $d \geq 3$ there are none. We formalize a finite instance.

Definition 76 (Cabello’s set^{Lean}). Cabello, Estebaranz, and García-Alcaine [Cabello et al. 1996] give 18 vectors of $\mathbb{R}^4$ with coordinates in $\{0, \pm 1\}$, and 9 sets of 4 of them:

$\{0001, 0010, 1100, \bar{1}\bar{1}00\}$, $\{0001, 0100, 1010, 10\bar{1}0\}$, $\{1\bar{1}\bar{1}\bar{1}, \bar{1}\bar{1}\bar{1}\bar{1}, 1100, 0011\}$,
 $\{1\bar{1}\bar{1}\bar{1}, 1111, 10\bar{1}0, 010\bar{1}\}$, $\{0010, 0100, 1001, 100\bar{1}\}$, $\{1\bar{1}\bar{1}\bar{1}, 1111, 100\bar{1}, 0110\}$,
 $\{1\bar{1}\bar{1}\bar{1}, 111\bar{1}, \bar{1}\bar{1}00, 0011\}$, $\{1\bar{1}\bar{1}\bar{1}, \bar{1}111, 1010, 010\bar{1}\}$, $\{111\bar{1}, \bar{1}111, 100\bar{1}, 0110\}$

where $\bar{1}$ stands for -1 .

Theorem 77 (Kochen–Specker for Cabello’s set^{Lean}). *Each of the 9 sets is an orthogonal basis of $\mathbb{R}^4$, and each of the 18 vectors lies in exactly two of them. No function from the 18 vectors to $\{0, 1\}$ takes the value 1 at exactly one vector of every basis.*

The first two statements are checked by evaluation. The third is a parity count. Summing such a function over the 9 bases gives 9, and counts every vector twice, so the sum is even.

Theorem 78 (no certain states^{Lean}). *For the presentation whose events are the 18 vectors and whose certain states are the $\{0, 1\}$ -valued functions summing to 1 on every basis, the set of certain states and Mix of it are empty. For every $\psi \in \mathbb{R}^4$ with $\psi \cdot \psi = 1$, the Born-rule assignment sums to 1 on every basis, so it is a quantum state that is not in Mix of the certain states.*

The Born-rule statement is Parseval’s identity $\sum_{v \in B} (\psi \cdot v)^2 / (v \cdot v) = \psi \cdot \psi$ for each orthogonal basis B , checked basis by basis^{Lean}.

Quantum probability therefore fails criterion (1) of Section 1: the calculus has no $\{0, 1\}$ -valued elements, so they cannot reconstruct the event logic. Theorem 78 proves this for Cabello’s set. For all rays of $\mathbb{R}^d$ with $d \geq 3$ it follows from the Kochen–Specker theorem, which is not formalized. The construction of Section 10 produces nothing at this event logic: with no certain states, every affine law is valid on the certain states, including $1 \leq 0$, and Theorem 63 returns the empty set. The mixtures of certain states are the noncontextual hidden-variable models of the event logic, and Theorem 77 is the statement that this event logic has none [Kochen and Specker 1967]. For lattice valuations, Theorem 55 shows the corresponding absence on the non-distributive lattice M_3 , which has no sharp valuation.

13 Related Work

Informal precedents. Pairing probability with intuitionistic logic is not a new idea, though the literature on it is scattered. Weatherson develops intuitionistic probability axiomatically and defends it with Dutch-book arguments [Weatherson 2003]. His axioms match our valuation

axioms exactly, with negation derived rather than assumed, so the object itself and the bound $v(a) + v(\neg a) \leq 1$ are his. What is new here is the decomposition of that bound’s slack into dnGap and deMorganGap and its determinacy from v alone (Theorem 6). Paris gives an informal theorem that any modular assignment of non-classical truth values is exactly a Dempster–Shafer belief function [Paris 1994]. We mechanize the analogous fact for our intuitionistic setting: the 2-monotone case follows directly from modularity^{Lean}, and the full tower is Theorem 23. The identification $\text{Bel} = \text{P}(\Box \cdot)$ connecting belief functions [Dempster 1967; Shafer 1976] to modal logic is established [Ruspini 1986]. Section 7 shows this identification is not a separate axiom for us but a special case of our frame-level slack, recovered automatically once the frame is a topology and $\neg$ is read as the interior of the complement^{Lean}.

On the localic side, valuations on frames are standard objects of constructive measure theory, with Vickers’s work and the Coquand–Spitters development as canonical references [Coquand and Spitters 2009; Vickers 2008]. These normally build in Scott continuity and target additive, Riesz-style goals. Section 4 shows Scott continuity to be a substantive assumption (it decides point-representability on the frames we examine), so our base structure omits it. Cox, Jaynes, Halpern’s counterexample, and the Van Horn and Arnborg–Sjödín repairs [Arnborg and Sjödín 2000; Cox 1946; Halpern 1999; Horn 2003; Jaynes 2003] are classical throughout, and Cox’s and Van Horn’s derivations both lean on double-negation elimination. Section 6 proves the sum-rule half without that move, at the price of positing modularity as an axiom rather than deriving it (Example 29). Van Horn observes that dropping axiom R3 yields an interval-valued theory, exactly the belief/plausibility pair our slack separates. To our knowledge, no prior work, informal or formal, states either characterization theorem: complement rule $\iff$ excluded middle (Theorem 10), or Dempster = Bayes $\iff$ zero slack at the evidence (Theorem 27).

Prior formalizations. Two bodies of mechanized work are adjacent, and neither overlaps. ALEA [Audebaud and Paulin-Mohring 2009] formalizes a discrete distribution monad in Coq for randomized programs: additive probability, no event logic in our sense. Closest is the HoTT/Coq line of Faissole and Spitters on *valuations* over lower reals [Faissole and Spitters 2017], done constructively with Scott continuity and aimed at integration (Riesz) and a Giry-style valuation *monad*. Their valuations differ from ours in carrying Scott continuity, and their monad is the Scott-continuous instance of the question examined in Section 8. Our development drops the continuity their base structure builds in, and the two mechanizations do not otherwise overlap. On the domain-theoretic side, commutativity of the full valuations monad on the category of directed-complete partial orders is open. A hierarchy of commutative submonads,

generated from simple valuations (finite mixtures of point valuations), has been constructed instead [Jia et al. 2021]. Our countably-presented fragment, the valuations given as countable mixtures (Proposition 49), is the frame-side analogue.

Non-Boolean probability elsewhere. Quantum probability on orthomodular lattices, with Gleason’s theorem [Birkhoff and von Neumann 1936; Gleason 1957], is the long-standing precedent for re-basing probability on a non-Boolean event logic. Orthomodular lattices give up distributivity itself, while our frames stay distributive by definition (Definition 1), locating all the non-classical behavior in the failure of excluded middle alone.

Linear logic marks a different kind of departure. Its event algebras are quantales, complete lattices equipped with an associative tensor $\otimes$ that need not be idempotent [Girard 1987]. The lattice reduct of a quantale is already an instance of Section 9’s generalization, but $\otimes$ itself is not $\sqcap$, and a valuation grading it cannot be phrased in Definition 50’s vocabulary without first deciding what modularity even means for a non-idempotent operation. The probabilistic semantics that does exist for linear logic, Girard’s probabilistic coherence spaces as developed by Danos and Ehrhard [Danos and Ehrhard 2011], grades the multiplicative structure directly rather than valuing a lattice, and is not a special case of the construction studied here.

Free probability, fuzzy measures, and imprecise probability [Walley 1991] vary other parameters (independence, additivity, single-valuedness). Where Walley’s coherent lower previsions posit the belief/plausibility gap directly as a coherence primitive, our slack (Section 2) is derived, forced by modularity and negation rather than assumed. Fritz’s synthetic Markov-category framework [Fritz 2020] varies the ambient structure categorically rather than order-theoretically, axiomatizing conditional independence and sufficient statistics abstractly across discrete, measure-theoretic, and Gaussian probability alike. It remains unmechanized and is orthogonal to our variation in the underlying logic. Wolpert and Kinney model mathematics itself as a stochastic process, replacing binary theoremhood with a probability distribution [Wolpert and Kinney 2024]. Theoremhood is a semidecidable event, so the computability guard of Section 7 bears directly on that program: sharp credences about such events are uncomputable. Our variation is specifically the *logic*, holding the Cox-style methodology fixed. Within DS theory, the two conditionings and their inequivalence are classical [Fagin and Halpern 1991; Shafer 1976]. What is new in Theorem 27 is the exact logical characterization of their agreement, and its mechanization.

14 Conclusion

In this work, we take seriously the Cox–Jaynes reading of probability as *extended logic*, and make the extension a construction with the logic of events as a parameter. The construction fixes an event algebra (the chosen logic in algebraic form) and takes as its assignments of degrees of certainty the valuations: the maps from events to numbers satisfying monotonicity, normalization, and modularity. We specify the two conditions that qualify the result as a probability calculus: when events are certain (0 or 1 certainty), the valuations must reconstruct the underlying event logic, and the value of the construction at classical logic must be the classical theory.

The motivation for developing this extension comes from the study of program behaviour in terms of outcome probabilities. Many properties of programs are only semidecidable. However, the Boolean event algebra underlying Kolmogorov’s axioms imposes the law of excluded middle (and therefore decidability) on the structure of logical connectives applied to events. This makes it not a good fit for reasoning about events that are not decidable, such as whether an arbitrary program halts: insisting on a sharp classical readout for such an event does not decide it, but instead produces an object we show to be uncomputable, and a non-sharp classical credence assigns its remainder to non-termination, an event no finite observation confirms. The extension process we propose, when fixing its logic parameter to be intuitionistic, does not conflict with the underlying event logic of computation by construction, and is capable of reasoning about the possibility of non-termination.

Substituting the Heyting event algebra (and therefore intuitionistic logic) for the Boolean algebra representing the classical logic case changes only the consequences that excluded middle supplied. Both qualifying conditions hold at this parameter value (Sections 2 and 3), and classicality enters only through the complement rule (Theorem 10) and its dynamic counterpart, the agreement of Dempster and Bayesian conditioning (Theorem 27). The complement rule degrades to the inequality $v(a) + v(\neg a) \leq 1$. The deficit in this inequality, for which we re-purpose the term slack, is a new quantity: the value of the undecided region, given by the boundary mass in the measure model. Slack is submodular yet does not determine the valuation. The weaker logic also adds the two-gap decomposition of slack, the atomic/diffuse divide, and a belief-function reading through $\neg\neg$.

Everything else holds verbatim on frames: conditioning, the product rule, Bayes symmetry, disjoint additivity, total probability over genuine partitions, and inclusion–exclusion itself. On finite frames, every valuation is a classical mass function on the frame’s points. The product-rule half of Cox’s theorem is independent of the logic while the sum-rule half

must be posited, so the direct transcription of the classical Cox statement is unsatisfiable. Every result is machine-checked in Lean 4 over mathlib.

A recurring shape. Three of this paper’s negative results share a form worth naming: two individually reasonable demands on a valuation, jointly unsatisfiable, with the proof identifying which demand is forced to give rather than merely asserting that some construction fails. On M_3 (Theorem 55), sharpness and modularity together are unsatisfiable: a 2-valued readout cannot satisfy all three pairwise sum-to-one equations at once, and distributivity is the hypothesis that has to give. Grading a non-idempotent tensor (Corollary 59) is the same shape one level up, at the axiom itself: being graded by modularity’s compositional law and discriminating a from $a \otimes a$ cannot both hold, for any Ω and any $\otimes$, and probabilistic coherence spaces [Danos and Ehrhard 2011] are the witness that keeps discrimination and gives up gradability. Demanding a sharp, uniform, *computable* readout of a semidecidable event (Theorem 45) is the same trade-off again, one level further out: bivalence and computability cannot both hold, and the classical route keeps bivalence by fiat and gives up an algorithm. None of the three results says only that a construction fails; each names the exact pair of demands responsible and shows the failure is forced, not an artifact of how the construction happened to be set up.

Toward a monad. Section 8 lays the groundwork for a categorical exploration of the construction: Val is a mechanized functor on locales, and its countably-presented fragment composes as kernels, though whether that composition is canonical on valuations rather than their presentations remains open. A commutative monad structure, where it exists, would be the intuitionistic counterpart of the Giry monad and a step toward compositional semantics for probabilistic programs with semidecidable events. One candidate formulation exists: the space of Scott-continuous valuations is the classifying locale of a geometric theory [Coquand and Spitters 2009; Vickers 2008]. Deleting its continuity axiom still leaves a geometric theory, whose points, read classically, are exactly our valuations. Its multiplication, however, must integrate against a merely modular valuation, and the diffuse part of Example 18 is exactly what such integration fails to capture. We offer the formulation as a target, not a result. Beyond intuitionistic logic, no ambient category sending a logic to its probability calculus functorially is evident from our construction, and this question remains open.

Other open problems and future work. A natural extension of our mechanization effort involves completing the Aczél development, and mechanizing the Dutch-book theorem [Weatherson 2003]. Both require substantial additional effort and were deemed out of scope for this work. To extend our mechanization effort with preliminary monad results, we would quotient the countable presentations (Proposition 49)

into a Kleisli category of countably-presented kernels, and mechanize the classifying-locale formulation (which needs classifying locales of geometric theories, infrastructure mathlib currently lacks).

Some related mathematical problems remain open as well. The first is developing a representation of the diffuse part on non-spatial frames. The second is deriving modularity from a criterion more primitive than the mixture characterization (Theorem 17), knowing it cannot come from disjunction data alone (Example 29).

The construction method applies beyond intuitionistic logic: state the classical desiderata, delete or substitute the axioms the target logic no longer licenses, and check what survives, what splits, and what becomes unsatisfiable. Linear logic was the first case tried, and Section 9 closes it in the negative rather than leaving it open: substituting the quantale tensor $\otimes$ for $\sqcap$ in modularity is not merely difficult but unsatisfiable (Corollary 59), so no valuation in this paper’s sense can grade $\otimes$ ’s consumption of evidence without losing the very discrimination $\otimes$ exists to carry [Danos and Ehrhard 2011]. What remains open is the next question this raises, not this instance of it: whether some compositional law other than modularity transplanted verbatim could play our construction’s role for logics whose primitive connectives are not idempotent.

References

- János Aczél. 1966. *Lectures on Functional Equations and Their Applications*. Academic Press, New York, NY, USA.
- Reynald Affeldt, Cyril Cohen, and Damien Rouhling. 2018. Formalization Techniques for Asymptotic Reasoning in Classical Analysis. *Journal of Formalized Reasoning* 11, 1 (2018), 43–76.
- Stefan Arnborg and Gunnar Sjödin. 2000. Bayes Rules in Finite Models. In *European Conference on Artificial Intelligence (ECAI)*. IOS Press, Amsterdam, Netherlands, 571–575.
- Philippe Audebaud and Christine Paulin-Mohring. 2009. Proofs of Randomized Algorithms in Coq. *Science of Computer Programming* 74, 8 (2009), 568–589.
- Garrett Birkhoff. 1937. Rings of Sets. *Duke Mathematical Journal* 3, 3 (1937), 443–454.
- Garrett Birkhoff and John von Neumann. 1936. The Logic of Quantum Mechanics. *Annals of Mathematics* 37, 4 (1936), 823–843.
- Adán Cabello, José M. Estebaranz, and Guillermo García-Alcaine. 1996. Bell-Kochen–Specker Theorem: A Proof with 18 Vectors. *Physics Letters A* 212, 4 (1996), 183–187.
- Adán Cabello, Simone Severini, and Andreas Winter. 2014. Graph-Theoretic Approach to Quantum Correlations. *Physical Review Letters* 112, 4 (2014), 040401.
- Gregory J. Chaitin. 1975. A Theory of Program Size Formally Identical to Information Theory. *J. ACM* 22, 3 (1975), 329–340.
- Maria Chudnovsky, Neil Robertson, Paul Seymour, and Robin Thomas. 2006. The Strong Perfect Graph Theorem. *Annals of Mathematics* 164, 1 (2006), 51–229.
- Václav Chvátal. 1975. On Certain Polytopes Associated with Graphs. *Journal of Combinatorial Theory, Series B* 18, 2 (1975), 138–154.
- Thierry Coquand and Bas Spitters. 2009. Integrals and Valuations. *Journal of Logic and Analysis* 1, 3 (2009), 1–22.
- Richard T. Cox. 1946. Probability, Frequency and Reasonable Expectation. *American Journal of Physics* 14, 1 (1946), 1–13.

- Vincent Danos and Thomas Ehrhard. 2011. Probabilistic Coherence Spaces as a Model of Higher-Order Probabilistic Computation. *Information and Computation* 209, 6 (2011), 966–991.
- Leonardo de Moura and Sebastian Ullrich. 2021. The Lean 4 Theorem Prover and Programming Language. In *Conference on Automated Deduction (CADE)*. Springer, Cham, Switzerland, 625–635.
- Arthur P. Dempster. 1967. Upper and Lower Probabilities Induced by a Multivalued Mapping. *Annals of Mathematical Statistics* 38, 2 (1967), 325–339.
- Dieter Denneberg. 1994. *Non-Additive Measure and Integral*. Kluwer, Dordrecht, Netherlands.
- Martin H. Escardó. 2004. Synthetic Topology of Data Types and Classical Spaces. *Electronic Notes in Theoretical Computer Science* 87 (2004), 21–156.
- Ronald Fagin and Joseph Y. Halpern. 1991. A New Approach to Updating Beliefs. *Uncertainty in Artificial Intelligence* 6 (1991), 347–374.
- Florian Faissolle and Bas Spitters. 2017. Synthetic Topology in Homotopy Type Theory for Probabilistic Programming. In *Programming Languages meets Program Verification (PPS)*. Full version with M. E. Bidlingmaier: Faissolle–Bidlingmaier–Spitters, “Synthetic topology in Homotopy Type Theory for probabilistic programming,” *Math. Struct. Comput. Sci.* 31(4), 2021..
- Yannick Forster, Dominik Kirst, and Gert Smolka. 2019. On Synthetic Undecidability in Coq, with an Application to the Entscheidungsproblem. In *Proceedings of the 8th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP)*. 38–51.
- Tobias Fritz. 2020. A Synthetic Approach to Markov Kernels, Conditional Independence and Theorems on Sufficient Statistics. *Advances in Mathematics* 370 (2020), 107239.
- Paolo Ghirardato. 1997. On Independence for Non-Additive Measures, with a Fubini Theorem. *Journal of Economic Theory* 73, 2 (1997), 261–291.
- Jean-Yves Girard. 1987. Linear Logic. *Theoretical Computer Science* 50, 1 (1987), 1–101.
- Michèle Giry. 1982. A Categorical Approach to Probability Theory. In *Categorical Aspects of Topology and Analysis*. Lecture Notes in Mathematics, Vol. 915. Springer, Berlin, Germany, 68–85.
- Andrew M. Gleason. 1957. Measures on the Closed Subspaces of a Hilbert Space. *Journal of Mathematics and Mechanics* 6, 6 (1957), 885–893.
- Kurt Gödel. 1933. Eine Interpretation des intuitionistischen Aussagenkalküls. *Ergebnisse eines mathematischen Kolloquiums* 4 (1933), 39–40.
- Martin Grötschel, László Lovász, and Alexander Schrijver. 1988. *Geometric Algorithms and Combinatorial Optimization*. Springer.
- Joseph Y. Halpern. 1999. A Counterexample to Theorems of Cox and Fine. *Journal of Artificial Intelligence Research* 10 (1999), 67–85.
- Carl G. Hempel. 1945. Studies in the Logic of Confirmation (I.). *Mind* 54, 213 (1945), 1–26.
- Otto Hölder. 1901. Die Axiome der Quantität und die Lehre vom Mass. *Berichte über die Verhandlungen der Sächsischen Gesellschaft der Wissenschaften zu Leipzig* 53 (1901), 1–64.
- Kevin S. Van Horn. 2003. Constructing a Logic of Plausible Inference: A Guide to Cox’s Theorem. *International Journal of Approximate Reasoning* 34, 1 (2003), 3–24.
- Edwin T. Jaynes. 2003. *Probability Theory: The Logic of Science*. Cambridge University Press, Cambridge, UK.
- Xiaodong Jia, Bert Lindenhovius, Michael Mislove, and Vladimir Zamdzhiev. 2021. Commutative Monads for Probabilistic Programming Languages. In *Logic in Computer Science (LICS)*. IEEE.
- Peter T. Johnstone. 1982. *Stone Spaces*. Cambridge University Press, Cambridge, UK.
- Alexander A. Klyachko, M. Ali Can, Sinem Binicioğlu, and Alexander S. Shumovsky. 2008. Simple Test for Hidden Variables in Spin-1 Systems. *Physical Review Letters* 101, 2 (2008), 020403.
- Simon Kochen and Ernst P. Specker. 1967. The Problem of Hidden Variables in Quantum Mechanics. *Journal of Mathematics and Mechanics* 17, 1 (1967), 59–87.
- Anders Kock. 2012. Commutative Monads as a Theory of Distributions. *Theory and Applications of Categories* 26 (2012), 97–131.
- Andrey N. Kolmogorov. 1933. *Grundbegriffe der Wahrscheinlichkeitsrechnung*. Springer, Berlin, Germany.
- Harold W. Kuhn. 1953. Extensive Games and the Problem of Information. In *Contributions to the Theory of Games II*. Annals of Mathematics Studies, Vol. 28. Princeton University Press, 193–216.
- László Lovász. 1972. Normal Hypergraphs and the Perfect Graph Conjecture. *Discrete Mathematics* 2, 3 (1972), 253–267.
- László Lovász. 1979. On the Shannon Capacity of a Graph. *IEEE Transactions on Information Theory* 25, 1 (1979), 1–7.
- Assia Mahboubi and Enrico Tassi. 2021. *Mathematical Components*. Zenodo. <https://doi.org/10.5281/zenodo.4457887>
- The mathlib Community. 2020. The Lean Mathematical Library. In *Certified Programs and Proofs (CPP)*. ACM, New York, NY, USA, 367–381.
- J. C. C. McKinsey and Alfred Tarski. 1944. The Algebra of Topology. *Annals of Mathematics* 45, 1 (1944), 141–191.
- Jeff B. Paris. 1994. *The Uncertain Reasoner’s Companion: A Mathematical Perspective*. Cambridge University Press, Cambridge, UK.
- K. P. S. Bhaskara Rao and M. Bhaskara Rao. 1983. *Theory of Charges: A Study of Finitely Additive Measures*. Pure and Applied Mathematics, Vol. 109. Academic Press, New York, NY, USA.
- Giuseppe Rosolini. 1986. *Continuity and Effectiveness in Topoi*. Ph. D. Dissertation. University of Oxford.
- Enrique H. Ruspini. 1986. *The Logical Foundations of Evidential Reasoning*. Technical Report 408. SRI International, Menlo Park, CA, USA.
- Glenn Shafer. 1976. *A Mathematical Theory of Evidence*. Princeton University Press, Princeton, NJ, USA.
- Steven Vickers. 2008. A Localic Theory of Lower and Upper Integrals. *Mathematical Logic Quarterly* 54, 1 (2008), 109–123.
- Peter Walley. 1991. *Statistical Reasoning with Imprecise Probabilities*. Chapman and Hall, London, UK.
- Brian Weatherston. 2003. From Classical to Intuitionistic Probability. *Notre Dame Journal of Formal Logic* 44, 2 (2003), 111–123.
- David H. Wolpert and David B. Kinney. 2024. A Stochastic Model of Mathematics and Science. *Foundations of Physics* 54 (2024), 21.
- Adam Yedidia and Scott Aaronson. 2016. A Relatively Small Turing Machine Whose Behavior Is Independent of Set Theory. *Complex Systems* 25, 4 (2016), 297–327.

A Values in the Extended Reals

Valuations take values in the extended non-negative reals $[0, \infty]$, following the measure-theoretic convention of mathlib. As a result, sums indexed by an arbitrary family are unconditionally defined, with no summability hypothesis, and they commute over dependent sums: for a function f on a sigma type $\Sigma a, \beta(a)$, one has $\sum_p f(p) = \sum_a \sum_b f(a, b)$ with no side condition. This is what lets the countable-mixture results of Section 8 hold as stated. Additionally, subtraction is truncated at 0. This never affects the main text, since every subtraction there is applied to a pair already ordered by monotonicity, as with the mass of Section 4, so the truncation is never triggered.

B Lean Statements

B.1 Introduction

B.2 Valuations on Frames

```

2421 structure Valuation ( $\Omega$  : Type*) [Order.
2422   Frame  $\Omega$ ] where
2423   toFun :  $\Omega \rightarrow \mathbb{R}_{\geq 0}$ 
2424   map_bot' : toFun  $\perp$  = 0
2425   map_top' : toFun  $\top$  = 1
2426   mono' : Monotone toFun
2427   modular' :  $\forall a b$ , toFun  $a + \text{toFun } b =$ 
2428     toFun  $(a \sqcup b) + \text{toFun } (a \sqcap b)$ 
2429
2430 theorem Valuation.add_compl_le_one ( $v$  :
2431   Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :
2432    $v a + v a^c \leq 1$ 
2433
2434 theorem Valuation.add_compl_eq_sup ( $v$  :
2435   Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :
2436    $v a + v a^c = v (a \sqcup a^c)$ 
2437
2438 theorem Valuation.le_one ( $v$  : Valuation  $\Omega$ )
2439   ( $a$  :  $\Omega$ ) :  $v a \leq 1$ 
2440
2441 theorem Valuation.
2442   slack_eq_dnGap_add_deMorganGap ( $v$  :
2443     Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :
2444      $v.\text{slack } a = v.\text{dnGap } a + v.\text{deMorganGap}$ 
2445        $a$ 
2446
2447 theorem Valuation.slack_eq_sub_add_sub ( $v$ 
2448   : Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :
2449      $v.\text{slack } a = (v a^{cc} - v a) + (1 - (v a$ 
2450        $^{cc} + v a^c))$ 
2451
2452 theorem Valuation.le_compl_compl ( $v$  :
2453   Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :  $v a \leq v a^{cc}$ 
2454
2455 theorem Valuation.slack_submodular ( $v$  :
2456   Valuation  $\Omega$ ) ( $a b$  :  $\Omega$ ) :
2457      $v.\text{slack } (a \sqcup b) + v.\text{slack } (a \sqcap b) \leq v.$ 
2458        $\text{slack } a + v.\text{slack } b$ 
2459
2460 def Order.Frame.IsDeMorgan ( $\Omega$  : Type*) [
2461   Order.Frame  $\Omega$ ] : Prop :=
2462    $\forall a : \Omega$ ,  $a^{cc} \sqcup a^c = \top$ 
2463
2464 theorem Valuation.
2465   slack_eq_dnGap_of_isDeMorgan
2466   ( $h\Omega$  : Order.Frame.IsDeMorgan  $\Omega$ ) ( $v$  :
2467     Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :
2468      $v.\text{slack } a = v.\text{dnGap } a$ 
2469
2470 noncomputable def Valuation.condVal ( $v$  :
2471   Valuation  $\Omega$ ) ( $b$  :  $\Omega$ ) ( $hb$  :  $v b \neq 0$ ) :
2472   Valuation  $\Omega$ 

```

```

2476 theorem Valuation.condVal_mul ( $v$  :
2477   Valuation  $\Omega$ ) ( $b$  :  $\Omega$ ) ( $hb$  :  $v b \neq 0$ ) ( $a$ 
2478   :  $\Omega$ ) :
2479      $v.\text{condVal } b hb a * v b = v (a \sqcap b)$ 
2480
2481 theorem Valuation.condVal_symm ( $v$  :
2482   Valuation  $\Omega$ ) { $a b$  :  $\Omega$ }
2483   ( $ha$  :  $v a \neq 0$ ) ( $hb$  :  $v b \neq 0$ ) :
2484      $v.\text{condVal } b hb a * v b = v.\text{condVal } a$ 
2485        $ha b * v a$ 
2486
2487 theorem Valuation.cond_add_compl_le ( $v$  :
2488   Valuation  $\Omega$ ) ( $a b$  :  $\Omega$ ) :
2489      $v (a \sqcap b) + v (a^c \sqcap b) \leq v b$ 
2490
2491 theorem Valuation.
2492   add_compl_eq_one_of_complemented ( $v$  :
2493     Valuation  $\Omega$ ) { $a$  :  $\Omega$ }
2494     ( $ha$  :  $a \sqcup a^c = \top$ ) :  $v a + v a^c = 1$ 
2495
2496 variable [CompleteBooleanAlgebra  $\Omega$ ]
2497
2498 theorem classical_additivity ( $v$  :
2499   Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :
2500      $v a + v a^c = 1$ 
2501
2502 variable [CompleteBooleanAlgebra  $\Omega$ ]
2503
2504 theorem classical_slack_zero ( $v$  :
2505   Valuation  $\Omega$ ) ( $a$  :  $\Omega$ ) :
2506      $v.\text{slack } a = 0$ 
2507
2508 noncomputable def Valuation.mix ( $w$  :  $\iota \rightarrow \mathbb{R}$ 
2509    $\geq 0$ ) ( $hw$  :  $\sum i, w i = 1$ )
2510   ( $v$  :  $\iota \rightarrow$  Valuation  $\Omega$ ) : Valuation  $\Omega$ 
2511
2512 B.3 Where Classicality Lives
2513
2514 def Valuation.HasClassicalNegation ( $v$  :
2515   Valuation  $\Omega$ ) : Prop :=
2516    $\forall a$ ,  $v a + v a^c = 1$ 
2517
2518 theorem hasClassicalNegation_of_em ( $hem$  :
2519    $\forall a : \Omega$ ,  $a \sqcup a^c = \top$ )
2520   ( $v$  : Valuation  $\Omega$ ) :  $v.$ 
2521     HasClassicalNegation
2522
2523 theorem em_of_forall_hasClassicalNegation
2524   ( $h$  :  $\forall v$  : Valuation  $\Omega$ ,  $v.$ 
2525     HasClassicalNegation) :  $\forall a : \Omega$ ,  $a$ 
2526      $\sqcup a^c = \top$ 
2527
2528 theorem sharp_iff_point ( $v$  : Valuation  $\Omega$ )
2529   :
2530

```

```

2531   v.IsSharp  $\leftrightarrow \exists (J : \text{Order.Ideal } \Omega) (hJ$ 
2532       : J.IsPrime) (htop :  $\top \notin J$ ),
2533       v = Ideal.toValuation J hJ htop
2534
2535   theorem
2536     sharp_scott_iff_completelyPrimePoint (v
2537       : Valuation  $\Omega$ ) :
2538     v.IsSharp  $\wedge$  v.ScottContinuous  $\leftrightarrow$ 
2539      $\exists (J : \text{Order.Ideal } \Omega) (hJ : \text{Ideal.}$ 
2540       IsCompletelyPrime J) (htop :  $\top \notin$ 
2541       J),
2542     v = Ideal.toValuation J hJ.1 htop
2543
2544
2545
2546

```

B.4 Valuation Representation

```

2547   noncomputable def Valuation.mass (v :
2548     Valuation (LowerSet P)) (p : P) :  $\mathbb{R}_{\geq 0\infty}$ 
2549
2550   theorem Valuation.eq_sum_mass (v :
2551     Valuation (LowerSet P)) (U : LowerSet P
2552     ) :
2553     v U =  $\sum p \in U.$ toFinset, v.mass p
2554
2555   theorem Valuation.sum_mass (v : Valuation
2556     (LowerSet P)) :  $\sum p, v.$ mass p = 1
2557
2558   noncomputable def Valuation.toPMF (v :
2559     Valuation (LowerSet P)) : PMF P
2560
2561   theorem Valuation.exists_pmf_presentation
2562     { $\Theta : \text{Type } u$ } [Order.Frame  $\Theta$ ] [Finite  $\Theta$ ]
2563     (v : Valuation  $\Theta$ ) :
2564      $\exists (P : \text{Type } u) (\_ : \text{PartialOrder } P) (\_$ 
2565       : Fintype P) ( $\_ : \text{DecidableEq } P$ )
2566       (e :  $\Theta \approx_o \text{LowerSet } P$ ),
2567       ( $\forall a, v a = \sum p \in (e a).$ toFinset, (v
2568         .mapEquiv e).mass p)  $\wedge$ 
2569        $\sum p, (v.$ mapEquiv e).mass p = 1
2570
2571   noncomputable def deltaPoint (p : P) :
2572     Valuation (LowerSet P)
2573
2574   theorem Valuation.eq_mix_deltaPoint (v :
2575     Valuation (LowerSet P)) :
2576     v = Valuation.mix v.mass v.sum_mass (
2577       fun p => deltaPoint p)
2578
2579   theorem
2580     exists_valuation_not_point_representable
2581     :
2582      $\exists v : \text{Valuation } (\text{LowerSet } \mathbb{N}), \sum' p, v.$ 
2583       mass p  $\neq \top$ 
2584
2585

```

```

2586   theorem topIndicator_mass (n :  $\mathbb{N}$ ) :
2587     topIndicator.mass n = 0
2588
2589   theorem topIndicator_not_scott :
2590     ( $\sqcup n, \text{topIndicator } (\text{LowerSet.Iic } n)$ )  $\neq$ 
2591     topIndicator  $\top$ 
2592
2593   theorem eq_tsum_mass_of_scott (v :
2594     Valuation (LowerSet  $\mathbb{N}$ ))
2595     (hsc :  $v \top = \sqcup n, v (\text{LowerSet.Iic } n)$ )
2596     :  $v \top = \sum' n, v.$ mass n
2597
2598   variable [ $\forall p : P, \text{Finite } (\text{Set.Iic } p)$ ]
2599
2600   theorem Valuation.isPurelyAtomic_of_scott
2601     (v : Valuation (LowerSet P))
2602     (hsc :  $v \top = \sqcup (S : \text{Finset } P), v (S.$ 
2603       sup LowerSet.Iic)) : IsPurelyAtomic
2604       v
2605
2606   theorem tsum_mass_le (v : Valuation (
2607     LowerSet P)) :  $\sum' p, v.$ mass p  $\leq v \top$ 
2608
2609

```

B.5 The Dempster–Shafer Reading

```

2610   theorem Valuation.inclusion_exclusion (v :
2611     Valuation  $\Omega$ ) (s : Finset  $\iota$ )
2612     (a :  $\iota \rightarrow \Omega$ ) : v (s.sup a) + v.iesEven
2613       s a = v.iesOdd s a
2614
2615   theorem Valuation.infty_monotone (v :
2616     Valuation  $\Omega$ ) (s : Finset  $\iota$ ) (a :  $\iota \rightarrow \Omega$ )
2617     :
2618     v.iesOdd s a  $\leq v ((s.\text{sup } a)^{cc}) + v.$ 
2619       iesEven s a
2620
2621   theorem Valuation.two_monotone (v :
2622     Valuation  $\Omega$ ) (a b :  $\Omega$ ) :
2623     v a + v b  $\leq v (a \sqcup b)^{cc} + v (a \sqcap b)$ 
2624
2625   noncomputable def Valuation.plausibility (
2626     v : Valuation  $\Omega$ ) (a :  $\Omega$ ) :  $\mathbb{R}_{\geq 0\infty} := 1 -$ 
2627     v ac
2628
2629   theorem Valuation.plausibility_sub_self (v :
2630     Valuation  $\Omega$ ) (a :  $\Omega$ ) :
2631     v.plausibility a - v a = v.slack a
2632
2633   theorem Valuation.sup_compl_decomp (v :
2634     Valuation  $\Omega$ ) (a b :  $\Omega$ ) :
2635     v (a  $\sqcup$  bc) = v bc + (v (a  $\sqcap$  b) + v.
2636       emGap a b)
2637
2638

```

```

2641 noncomputable def Valuation.dempsterCond (
2642   v : Valuation Ω) (b : Ω)
2643   (hb : v b ≠ 0) : Valuation Ω
2644
2645 theorem Valuation.dempsterCond_apply (v :
2646   Valuation Ω) (b : Ω) (hb : v b ≠ 0)
2647   (a : Ω) : v.dempsterCond b hb a = (v (
2648     a ⊔ bc) - v bc) / (1 - v bc)
2649
2650 theorem Valuation.
2651   dempsterCond_eq_condVal_iff_slack (v :
2652     Valuation Ω) {b : Ω}
2653   (hb : v b ≠ 0) :
2654   (∀ a, v.dempsterCond b hb a = v.
2655     condVal b hb a) ↔ v.slack b = 0
2656
2657 B.6 Cox's Theorem
2658
2659 structure CoxModel (Ω : Type*) [Order.
2660   Frame Ω] where
2661   pl : Ω → Ω → ℝ
2662   mono_left : ∀ c, Monotone fun a => pl a
2663     c
2664   pl_bot : ∀ c, pl ⊥ c = 0
2665   pl_top : ∀ c, pl ⊤ c = 1
2666   F : ℝ → ℝ → ℝ
2667   conj : ∀ a b c, pl (a ⊓ b) c = F (pl a (
2668     b ⊓ c)) (pl b c)
2669   F_assoc : ∀ x y z, F (F x y) z = F x (F
2670     y z)
2671   F_strictMono_left : ∀ y, 0 < y →
2672     StrictMono fun x => F x y
2673
2674 theorem isDeMorgan_lowerSetV : Order.Frame
2675   .IsDeMorgan (LowerSet V)
2676
2677 theorem modularity_irreducible :
2678   ∃ (Ω : Type) (⊥ : Order.Frame Ω) (q :
2679     Ω → ℝ≥0),
2680   Monotone q ∧ q ⊥ = 0 ∧ q ⊤ = 1 ∧
2681   (∀ x y : Ω, x ⊓ y = ⊥ → q (x ⊔ y) =
2682     q x + q y) ∧
2683   ¬ (∀ x y : Ω, q x + q y = q (x ⊔ y)
2684     + q (x ⊓ y))
2685
2686 theorem no_disjunction_functional :
2687   ¬ ∃ S : ℝ≥0 → ℝ≥0 → ℝ≥0,
2688   ∀ x y : LowerSet V, qV (x ⊔ y) = S
2689     (qV x) (qV y)
2690
2691 theorem constructive_cox (M : CoxModel Ω)
2692   (hmod : ∀ a b : Ω, M.pl a ⊤ + M.pl b ⊤
2693     = M.pl (a ⊔ b) ⊤ + M.pl (a ⊓ b) ⊤)
2694   :
2695

```

```

2696   ∃ (v : Valuation Ω) (g : ℝ → ℝ≥0),
2697   StrictMonoOn g (Set.Icc 0 1) ∧ g 0 =
2698     0 ∧ g 1 = 1 ∧
2699   ∀ a, v a = g (M.pl a ⊤)
2700
2701 noncomputable def coxModelENNReal :
2702   CoxModel ℝ≥0
2703
2704 theorem Scale.exists_mul_generator :
2705   ∃ G : ℝ → ℝ,
2706   (∀ x y, S.u ≤ x → S.u ≤ y → G (S.F x
2707     y) = G x * G y) ∧
2708   (∀ x y, S.u ≤ x → x < y → G x < G y)
2709
2710 theorem Scale.aczelStatement_cone :
2711   ∃ G : ℝ → ℝ,
2712   (∀ x y, S.u ≤ x → S.u ≤ y → G (S.F x
2713     y) = G x * G y) ∧
2714   StrictMonoOn G (Set.Ici S.u)
2715
2716 theorem exists_diag_eq {F : ℝ → ℝ → ℝ} {a
2717   b c : ℝ} (hab : a ≤ b)
2718   (hcont : ContinuousOn (fun x => F x x)
2719     (Set.Icc a b))
2720   (hc : c ∈ Set.Icc (F a a) (F b b)) : ∃
2721     x ∈ Set.Icc a b, F x x = c
2722
2723 structure Scale where
2724   F : ℝ → ℝ → ℝ
2725   hdiv : ∀ c, ∃ x, F x x = c
2726   hassoc : ∀ x y z, F (F x y) z = F x (F y
2727     z)
2728   hpos : ∀ c x, x < F x c
2729   hmono : ∀ x1 x2 y1 y2 : ℝ, x1 ≤ x2 → y1
2730     ≤ y2 → F x1 y1 ≤ F x2 y2
2731   hcont : Continuous fun p : ℝ × ℝ => F p
2732     .1 p.2
2733   hbot : ∀ x, Filter.Tendsto (fun z => F x
2734     z) Filter.atBot (nhds x)
2735   u : ℝ
2736

```

B.7 Two Grounding Models

```

2739 noncomputable def MeasureTheory.Measure.
2740   toValuationOpens
2741   (μ : Measure X) [IsProbabilityMeasure
2742     μ] : Valuation (Opens X)
2743
2744 theorem toValuationOpens_slack_eq_frontier
2745   (μ : Measure X)
2746   [IsProbabilityMeasure μ] (U : Opens X)
2747   :
2748   μ.toValuationOpens.slack U = μ (
2749     frontier (U : Set X))
2750

```

2751	theorem classical_posterior_compl_eq (μ :	noncomputable def sharpReadout ($n : \mathbb{N}$) (c	2806
2752	Measure X)	: Code) : Valuation (Opens Prop) :=	2807
2753	[IsProbabilityMeasure μ] ($a\ b$: Opens	sierpinskiValuation (	2808
2754	X)	haltingWeight_le_one n c)	2809
2755	(hb : μ .toValuationOpens $b \neq 0$) :		2810
2756	$\mu ((a : \text{Set } X)^c \hat{a} \otimes b) / \mu\ b$	theorem sharpReadout_not_computable ($n : \mathbb{N}$	2811
2757	$= \mu$.toValuationOpens.condVal $b\ hb\ a^c$	) :	2812
2758	$+ \mu$ (frontier ($a : \text{Set } X$) $\hat{a} \otimes b$) /	$\neg$ ComputablePred fun $c : \text{Code} \Rightarrow$	2813
2759	$\mu\ b$	sharpReadout n c haltsOpen = 1	2814
2760			2815
2761	theorem interiorMeasure_add_compl_le (μ :	theorem isOpen_wordCylinder ($w : \text{List Bool}$	2816
2762	Measure X) [IsProbabilityMeasure μ]	) : IsOpen (wordCylinder w)	2817
2763	($S : \text{Set } X$) : μ .interiorMeasure S + μ .		2818
2764	interiorMeasure $S^c \leq 1$	theorem cantorMeasure_wordCylinder (w :	2819
2765		List Bool) :	2820
2766	theorem	cantorMeasure (wordCylinder w) = (1 /	2821
2767	isDeMorgan_opens_iff_extremallyDisconnected	2) ^ w.length	2822
2768	:		2823
2769	Order.Frame.IsDeMorgan (Opens X) $\leftrightarrow$	theorem	2824
2770	ExtremallyDisconnected X	disjoint_wordCylinder_of_not_prefix {v	2825
2771		w : List Bool}	2826
2772	theorem isDeMorgan_opens_iff_stonean [	(hv : $\neg v <+ w$) (hw : $\neg w <+ v$) :	2827
2773	CompactSpace X] [T2Space X] :	Disjoint (wordCylinder v) (	2828
2774	Order.Frame.IsDeMorgan (Opens X) $\leftrightarrow$	wordCylinder w)	2829
2775	ExtremallyDisconnected X		2830
2776		theorem cantorMeasure_biUnion_wordCylinder	2831
2777	inductive ThreePt : Type	{F : Finset (List Bool)}	2832
2778	ptA : ThreePt	(hF : PrefixFree F) :	2833
2779	ptB : ThreePt	cantorMeasure ($\bigcup w \in F$, wordCylinder w	2834
2780	ptZ : ThreePt	) = $\sum w \in F$, (1 / 2 : $\mathbb{R}_{\geq 0\infty}$) ^ w.	2835
2781		length	2836
2782	theorem frontier_subset_ptZ {U : Set		2837
2783	ThreePt} (hU : IsOpen U) :		2838
2784	frontier U $\subseteq \{\text{ptZ}\}$		2839
2785		B.8 The Monad Question	2840
2786	theorem exists_ne_valuation_same_slack :	def Valuation.comap ($f : \text{FrameHom } \Gamma\ \Omega$) (v	2841
2787	$\exists v\ v' : \text{Valuation (Opens ThreePt)}$, v.	: Valuation Ω) : Valuation Γ	2842
2788	slack = v'.slack $\wedge v \neq v'$		2843
2789		theorem Valuation.comap_id ($v : \text{Valuation}$	2844
2790	theorem haltsOpen_compl : haltsOpen ^c = ($\perp$	Ω) : v.comap (FrameHom.id Ω) = v	2845
2791	: Opens Prop)		2846
2792		theorem Valuation.comap_comp ($f : \text{FrameHom}$	2847
2793	theorem slack_eq_boundary {p : $\mathbb{R}_{\geq 0\infty}$ } ($hp1$	$\Gamma\ \Omega$) ($g : \text{FrameHom } \Delta\ \Gamma$)	2848
2794	: p ≤ 1) :	(v : Valuation Ω) : v.comap (f.comp g)	2849
2795	(sierpinskiValuation hp1).slack	= (v.comap f).comap g	2850
2796	haltsOpen		2851
2797	= coinMeasure p (frontier {q : Prop	theorem exists_rect_decomposition (W :	2852
2798	q})	LowerSet ($\mathbb{N} \times \mathbb{N}$)) :	2853
2799		$\exists (N : \mathbb{N}) (A\ B : \mathbb{N} \rightarrow \text{LowerSet } \mathbb{N})$,	2854
2800	noncomputable def haltingWeight ($n : \mathbb{N}$) (c	W = (Finset.range N).sup fun i => A	2855
2801	: Code) : $\mathbb{R}_{\geq 0\infty} :=$	i $\times^5$ B i	2856
2802	if (c.eval n).Dom then 1 else 0		2857
2803		theorem Valuation.eq_on_finsetSup_of_eq_on	2858
2804		{G : Set Ω }	2859
2805			2860

```

2861 (hG : ∀ g ∈ G, ∀ g' ∈ G, g ⊔ g' ∈ G) (
2862   u u' : Valuation Ω)
2863 (h : ∀ g ∈ G, u g = u' g) {ι : Type*}
2864 [DecidableEq ι]
2865 (s : Finset ι) (a : ι → Ω) (ha : ∀ i ∈
2866   s, a i ∈ G) :
2867 u (s.sup a) = u' (s.sup a)
2868
2869 theorem Valuation.product_unique (v w :
2870   Valuation (LowerSet ℕ))
2871 {u u' : Valuation (LowerSet (ℕ × ℕ))}
2872 (hu : ∀ A B : LowerSet ℕ, u (A ×s B) =
2873   v A * w B)
2874 (hu' : ∀ A B : LowerSet ℕ, u' (A ×s B)
2875   = v A * w B) : u = u'
2876
2877 noncomputable def ultrafilterValuation (W
2878   : Ultrafilter ι) : Valuation (Set ι)
2879
2880 theorem rect_mem_iff {α β : Type*} {W :
2881   Ultrafilter (α × β)}
2882 {S : Set α} {T : Set β} :
2883 S ×s T ∈ W ↔ S ∈ W.map Prod.fst ∧ T ∈
2884   W.map Prod.snd
2885
2886 theorem product_valuation_not_unique :
2887   ∃ u u' : Valuation (Set (ℕ × ℕ)),
2888     (∀ S T : Set ℕ, u (S ×s T) = u' (S ×
2889       s T)) ∧ u ≠ u'
2890
2891 theorem fubini_fails_for_valuations :
2892   ∃ (v w : Valuation (Set ℕ)) (u u' :
2893     Valuation (Set (ℕ × ℕ))),
2894     (∀ S T : Set ℕ, u (S ×s T) = v S * w
2895       T) ∧
2896     (∀ S T : Set ℕ, u' (S ×s T) = v S *
2897       w T) ∧ u ≠ u'
2898
2899 noncomputable def Valuation.mixCountable {
2900   ι : Type*} (w : ι → ℝ≥0∞)
2901   (hw : ∑' i, w i = 1) (v : ι →
2902     Valuation Ω) : Valuation Ω
2903
2904 theorem Valuation.
2905   mixCountable_mixCountable {ι : Type*} {
2906     κ : ι → Type*}
2907   (w : ι → ℝ≥0∞) (hw : ∑' i, w i = 1)
2908   (u : ∀ i, κ i → ℝ≥0∞) (hu : ∀ i, ∑' j,
2909     u i j = 1)
2910   (v : ∀ i, κ i → Valuation Ω) :
2911     Valuation.mixCountable w hw
2912     (fun i => Valuation.mixCountable (
2913       u i) (hu i) (v i))
2914
2915

```

```

= Valuation.mixCountable (fun p : Σ
   i, κ i => w p.1 * u p.1 p.2)
   (tsum_sigma_mul_eq_one w hw u hu)
   (fun p => v p.1 p.2)
2920
2921 theorem Valuation.mixCountable_unique (v :
2922   Valuation Ω) :
2923   Valuation.mixCountable (fun _ : PUnit
2924     => 1) (by simp) (fun _ => v) = v
2925
2926 theorem Valuation.mixCountable_comm {ι κ :
2927   Type*}
2928   (w : ι → ℝ≥0∞) (hw : ∑' i, w i = 1) (u
2929     : κ → ℝ≥0∞) (hu : ∑' j, u j = 1)
2930   (v : ι → κ → Valuation Ω) :
2931     Valuation.mixCountable w hw (fun i =>
2932       Valuation.mixCountable u hu (v i))
2933     = Valuation.mixCountable u hu
2934       (fun j => Valuation.mixCountable
2935         w hw (fun i => v i j))
2936
2937

```

B.9 Valuations Beyond Frames

Theorem 52.

```

2940 theorem exists_sharp_separating {a b : Ω}
2941   (hab : ¬ a ≤ b) :
2942   ∃ v : Valuation Ω, v.IsSharp ∧ v a = 1
2943     ∧ v b = 0
2944
2945

```

Definition 50.

```

2946 structure LValuation (Ω : Type*) [Lattice
2947   Ω] [BoundedOrder Ω] where
2948   toFun : Ω → ℝ≥0∞
2949   map_bot' : toFun ⊥ = 0
2950   map_top' : toFun ⊤ = 1
2951   mono' : Monotone toFun
2952   modular' : ∀ a b, toFun a + toFun b
2953     = toFun (a ⊔ b) + toFun (a ⊓
2954       b)
2955
2956

```

Proposition 51.

```

2957 def LValuation.comap (f :
2958   BoundedLatticeHom Γ Ω) (v : LValuation
2959     Ω) :
2960     LValuation Γ
2961
2962 theorem LValuation.comap_id (v :
2963   LValuation Ω) :
2964   v.comap (BoundedLatticeHom.id Ω) = v
2965
2966 theorem LValuation.comap_comp (f :
2967   BoundedLatticeHom Γ Ω)
2968   (g : BoundedLatticeHom Δ Γ) (v :
2969     LValuation Ω) :
2970

```

```

2971     v.comap (f.comp g) = (v.comap f).comap
2972       g

```

Theorem 53.

```

2975 theorem LValuation.exists_sharp_separating
2976   [DistribLattice  $\Omega$ ]
2977   {a b :  $\Omega$ } (hab :  $\neg a \leq b$ ) :
2978    $\exists v : \text{LValuation } \Omega, v.\text{IsSharp} \wedge v\ a =$ 
2979      $1 \wedge v\ b = 0$ 

```

Definition 54.

```

2981 inductive M3 : Type
2982   | z | a | b | c | o
2983
2984 instance : Lattice M3 where
2985   -- z bottom, o top, a b c pairwise-
2986     incomparable atoms;
2987   -- any two distinct atoms join to o and
2988     meet to z
2989
2990 instance : BoundedOrder M3 where
2991   top := o
2992   bot := z
2993
2994 theorem M3.not_distributive :
2995    $\neg \forall x\ y\ w : M3, x \sqcap (y \sqcup w) = (x \sqcap y)$ 
2996      $\sqcup (x \sqcap w)$ 

```

Theorem 55.

```

3000 theorem M3.no_sharp_valuation (v :
3001   LValuation M3) :  $\neg v.\text{IsSharp}$ 

```

Definition 56.

```

3003 def IsTensorValuation (v :  $\Omega \rightarrow \mathbb{R}_{\geq 0\infty}$ ) (t :
3004    $\Omega \rightarrow \Omega \rightarrow \Omega$ ) : Prop :=
3005    $\forall a\ b, v\ a + v\ b = v\ (a \sqcup b) + v\ (t\ a\ b)$ 
3006

```

Theorem 57.

```

3008 theorem IsTensorValuation.collapse {v :  $\Omega$ 
3009    $\rightarrow \mathbb{R}_{\geq 0\infty}$ } {t :  $\Omega \rightarrow \Omega \rightarrow \Omega$ }
3010   (h : IsTensorValuation v t) {a :  $\Omega$ } (
3011     ha : v a  $\neq \top$ ) :
3012     v (t a a) = v a
3013

```

Example 58.

```

3015 theorem forced_constant_of_add_valuation {
3016   v :  $\mathbb{N} \rightarrow \mathbb{R}_{\geq 0\infty}$ }
3017   (h : IsTensorValuation v ( $\hat{\cdot} + \hat{\cdot}$ )) (
3018     hfin :  $\forall n, v\ n \neq \top$ ) (n :  $\mathbb{N}$ ) :
3019     v n = v 0
3020

```

Corollary 59.

```

3022
3023
3024
3025

```

```

3026 theorem
3027   not_isTensorValuation_of_discriminates
3028   {v :  $\Omega \rightarrow \mathbb{R}_{\geq 0\infty}$ }
3029   {t :  $\Omega \rightarrow \Omega \rightarrow \Omega$ } {a :  $\Omega$ } (ha : v a  $\neq \top$ )
3030   (hne : v (t a a)  $\neq v\ a$ ) :
3031    $\neg \text{IsTensorValuation } v\ t$ 
3032

```

B.10 Probability Calculi from Certain States

```

3035 def IsSharpVec (x :  $E \rightarrow \mathbb{R}_{\geq 0\infty}$ ) : Prop :=  $\forall$ 
3036   e, x e = 0  $\vee$  x e = 1
3037

```

```

3038 def Mix (S : Set ( $E \rightarrow \mathbb{R}_{\geq 0\infty}$ )) : Set ( $E \rightarrow \mathbb{R}_{\geq}$ 
3039    $0\infty$ ) :=
3040   {x |  $\exists (n : \mathbb{N}) (w : \text{Fin } n \rightarrow \mathbb{R}_{\geq 0\infty}) (s :$ 
3041      $\text{Fin } n \rightarrow E \rightarrow \mathbb{R}_{\geq 0\infty}),$ 
3042      $(\forall i, s\ i \in S) \wedge \sum i, w\ i = 1 \wedge x = \sum$ 
3043        $i, w\ i \hat{\cdot} s\ i\}$ 
3044

```

```

3045 theorem sharp_mem_Mix_iff {S : Set ( $E \rightarrow \mathbb{R}_{\geq}$ 
3046    $0\infty$ )} (hS :  $\forall s \in S, \text{IsSharpVec } s$ )
3047   {x :  $E \rightarrow \mathbb{R}_{\geq 0\infty}$ } (hx : IsSharpVec x) : x
3048    $\in \text{Mix } S \leftrightarrow x \in S$ 
3049

```

```

3050 theorem Mix_Mix (S : Set ( $E \rightarrow \mathbb{R}_{\geq 0\infty}$ )) : Mix
3051   (Mix S) = Mix S
3052

```

```

3053 structure AffineLaw (E : Type*) [Fintype E]
3054   where
3055     a :  $E \rightarrow \mathbb{R}_{\geq 0\infty}$ 
3056     b :  $E \rightarrow \mathbb{R}_{\geq 0\infty}$ 
3057     k :  $\mathbb{R}_{\geq 0\infty}$ 
3058     k' :  $\mathbb{R}_{\geq 0\infty}$ 
3059

```

```

3060 def AffineLaw.Holds [Fintype E] (L :
3061   AffineLaw E) (x :  $E \rightarrow \mathbb{R}_{\geq 0\infty}$ ) : Prop :=
3062    $\sum e, L.a\ e * x\ e + L.k \leq \sum e, L.b\ e * x$ 
3063      $e + L.k'$ 
3064

```

```

3065 theorem AffineLaw.holds_of_Mix [Fintype E]
3066   (L : AffineLaw E) {S : Set ( $E \rightarrow \mathbb{R}_{\geq 0\infty}$ )}
3067   (hL :  $\forall s \in S, L.\text{Holds } s$ ) {x :  $E \rightarrow \mathbb{R}_{\geq 0}$ 
3068      $\infty$ } (hx : x  $\in \text{Mix } S$ ) : L.Holds x
3069

```

```

3070 theorem Mix_eq_laws' {S : Set ( $E \rightarrow \mathbb{R}_{\geq 0\infty}$ )}
3071   (hS :  $\forall s \in S, \text{IsSharpVec } s$ ) :
3072    $\text{Mix } S = \{x \mid \forall L : \text{AffineLaw } E, (\forall s \in$ 
3073      $S, L.\text{Holds } s) \rightarrow L.\text{Holds } x\}$ 
3074

```

```

3075 theorem frame_recipe :
3076   Set.range (fun v : Valuation (LowerSet
3077     P) => v.toFun)
3078

```

```

3081     = Mix {f | ∃ v : Valuation (LowerSet
3082       P), v.IsSharp ∧ f = v.toFun}
3083
3084   theorem Valuation.eq_deltaPoint_of_isSharp
3085     {v : Valuation (LowerSet P)} (hv : v.
3086     IsSharp) :
3087     ∃ p, v = deltaPoint p
3088
3089   structure FinLaw (E : Type*) where
3090     F : Finset E
3091     a : E → ℝ≥0∞
3092     b : E → ℝ≥0∞
3093     k : ℝ≥0∞
3094     k' : ℝ≥0∞
3095
3096   def FinLaw.Holds (L : FinLaw E) (x : E → ℝ
3097     ≥0∞) : Prop :=
3098     ∑ e ∈ L.F, L.a e * x e + L.k ≤ ∑ e ∈ L.F
3099     , L.b e * x e + L.k'
3100
3101   def restr (F : Finset E) (x : E → ℝ≥0∞) :
3102     F → ℝ≥0∞ := fun e => x e
3103
3104   theorem laws_iff_local_Mix {S : Set (E → ℝ
3105     ≥0∞)} (hS : ∀ s ∈ S, IsSharpVec s)
3106     (x : E → ℝ≥0∞) :
3107     (∀ L : FinLaw E, (∀ s ∈ S, L.Holds s)
3108     → L.Holds x) ↔
3109     ∀ F : Finset E, restr F x ∈ Mix (
3110     restr F '' S)
3111
3112   def Cfg (Ev : Type) := Ev → Bool
3113
3114   def Good (S : Set (Ev → ℝ≥0∞)) (F : Finset
3115     Ev) (σ : Cfg Ev) : Prop :=
3116     ∃ s ∈ S, ∀ e : F, s e.1 = vec σ e.1
3117
3118   def atom (e : Ev) : Set (Cfg Ev) := {σ | σ
3119     e = true}
3120
3121   def bad (S : Set (Ev → ℝ≥0∞)) (F : Finset
3122     Ev) : Set (Cfg Ev) := {σ | ¬ Good S F σ
3123     }
3124
3125   theorem laws_iff_barycenter {S : Set (Ev →
3126     ℝ≥0∞)} (hS : ∀ s ∈ S, IsSharpVec s)
3127     (x : Ev → ℝ≥0∞) :
3128     (∀ L : FinLaw Ev, (∀ s ∈ S, L.Holds s)
3129     → L.Holds x) ↔
3130     ∃ μ : ProbabilityMeasure (Cfg Ev), (
3131     ∀ F, (μ : Measure (Cfg Ev)) (bad
3132     S F) = 0) ∧
3133     ∀ e, x e = (μ : Measure (Cfg Ev))
3134     (atom e)
3135

```

```

def flat (X : Type*) [Fintype X] : Set (X
→ ℝ≥0∞) := {x | ∑ a, x a ≤ 1}
3136
3137
3138
noncomputable def app (t : X × Y → ℝ≥0∞) (
x : X → ℝ≥0∞) : Y → ℝ≥0∞ :=
3139
3140
3141   fun b => ∑ a, x a * t (a, b)
3142
3143   def lin (X Y : Type*) [Fintype X] [Fintype
3144     Y] : Set (X × Y → ℝ≥0∞) :=
3145     {t | ∀ x ∈ flat X, app t x ∈ flat Y}
3146
3147   theorem mem_lin_iff [DecidableEq X] (t : X
3148     × Y → ℝ≥0∞) :
3149     t ∈ lin X Y ↔ ∀ a, ∑ b, t (a, b) ≤ 1
3150
3151   def orth (S : Set (E → ℝ≥0∞)) : Set (E → ℝ
3152     ≥0∞) := {y | ∀ x ∈ S, pairing x y ≤ 1}
3153
3154   structure IsPCS (P : Set (E → ℝ≥0∞)) :
3155     Prop where
3156     closed : orth (orth P) = P
3157     total : ∀ e, ∃ x ∈ P, x e ≠ 0
3158     bounded : ∀ e, ∃ M ≠ 0, ∀ x ∈ P, x e ≤ M
3159
3160   theorem isPCS_lin : IsPCS (lin X Y)
3161
3162   theorem isPCS_flat : IsPCS (flat X)
3163
3164   theorem isPCS_tensDual : IsPCS tensDual
3165
3166   theorem pcs_recipe {X Y : Type} [Fintype X]
3167     [Fintype Y] [DecidableEq X] :
3168     lin X Y = Mix (Set.range (det : (X →
3169     Option Y) → X × Y → ℝ≥0∞))
3170
3171   theorem sharp_lin_iff_clique {X Y : Type*}
3172     [Fintype X] [Fintype Y] [DecidableEq X]
3173     [DecidableEq Y]
3174     {t : X × Y → ℝ≥0∞} (h01 : ∀ p, t p = 0
3175     ∨ t p = 1) :
3176     t ∈ lin X Y ↔ IsClique (arrowC (flatC
3177     X) (flatC Y)) (t ⌞ = 1)
3178
3179   def tD (X Y X' Y' : Type) [Fintype X] [
3180     Fintype Y] [Fintype X'] [Fintype Y'] :
3181     Set ((X × Y) × (X' × Y') → ℝ≥0∞) :=
3182     {t | ∀ y ∈ lin X Y, ∀ z ∈ lin X' Y', ∑ p
3183     , ∑ q, y p * t (p, q) * z q ≤ 1}
3184
3185   def FTree (a : X) (cf : Y → X') (pq : (X ×
3186     Y) × (X' × Y')) : Prop :=
3187     pq.1.1 = a ∧ pq.2.1 = cf pq.1.2
3188
3189
3190

```

```

3191 theorem sharp_tD_iff [Nonempty X] [
3192   Nonempty X'] {z : (X × Y) × (X' × Y') →
3193   ℝ≥0∞}
3194   (h01 : ∀ pq, z pq = 0 ∨ z pq = 1) :
3195   z ∈ tD X Y X' Y' ↔
3196   (∃ a cf, ∀ pq, z pq = 1 → FTree a cf
3197     pq) ∨ (∃ c af, ∀ pq, z pq = 1 →
3198     GTree c af pq)
3199
3200 theorem sharp_mem_sequentialG [Nonempty X]
3201   [Nonempty X'] {z : (X × Y) × (X' × Y')
3202   → ℝ≥0∞}
3203   (hz : z ∈ tD X Y X' Y') (h01 : ∀ u, z
3204     u = 0 ∨ z u = 1) : z ∈ sequentialG
3205     X Y X' Y'
3206
3207 structure Core (A B C D : Type) [Fintype A
3208   ] [Fintype C] where
3209   μ : A → ℝ≥0∞
3210   κ : A → B → C → ℝ≥0∞
3211   ρ : A → B → C → D → ℝ≥0∞
3212   hκ : ∀ a b, ∑ c, κ a b c ≤ 1
3213   hρ : ∀ a b c d, ρ a b c d ≤ 1
3214
3215 def sequentialG (X Y X' Y' : Type) [
3216   Fintype X] [Fintype X'] : Set ((X × Y)
3217   × (X' × Y') → ℝ≥0∞) :=
3218   {t | ∃ (F : Core X Y X' Y') (G : Core X'
3219     Y' X Y), F.mass + G.mass ≤ 1 ∧
3220     t = fun pq => F.val pq.1.1 pq.1.2 pq
3221       .2.1 pq.2.2 + G.val pq.2.1 pq.2.2
3222       pq.1.1 pq.1.2}
3223
3224 theorem recipe_eq_sequentialG [Nonempty X]
3225   [Nonempty X'] :
3226   Mix {z | z ∈ tD X Y X' Y' ∧ ∀ u, z u =
3227     0 ∨ z u = 1} = sequentialG X Y X'
3228     Y'
3229
3230 theorem sequentialG_subset_Mix :
3231   sequentialG X Y X' Y' ⊆ Mix (sharpSeqG
3232     X Y X' Y')
3233
3234 theorem Mix_sequentialG : Mix (sequentialG
3235     X Y X' Y') = sequentialG X Y X' Y'
3236
3237 theorem pentE_not_mix (hx : Function.
3238   Injective ex) (hy : Function.Injective
3239   ey)
3240   (hx' : Function.Injective ex') (hy' :
3241     Function.Injective ey')
3242   {n : ℕ} (w : Fin n → ℝ≥0∞) (hw : ∑ i,
3243     w i = 1) (z : Fin n → (X × Y) × (X'
3244     × Y') → ℝ≥0∞)

```

```

3245   (hz : ∀ i, z i ∈ tD X Y X' Y') (h01 :
3246     ∀ i u, z i u = 0 ∨ z i u = 1) :
3247     pentE ex ey ex' ey' ≠ ∑ i, w i â¢ z i
3248
3249 theorem pentE_mem (hx : Function.Injective
3250   ex) (hy : Function.Injective ey)
3251   (hx' : Function.Injective ex') (hy' :
3252     Function.Injective ey') :
3253     pentE ex ey ex' ey' ∈ tD X Y X' Y'
3254
3255 theorem pent_clique_le_two : ∀ B ∈ pent.
3256   powerset,
3257   (∀ s ∈ B, ∀ t ∈ B, compat2 s t) → B.
3258     card ≤ 2
3259
3260 theorem pent_indep_le_two : ∀ B ∈ pent.
3261   powerset,
3262   (∀ s ∈ B, ∀ t ∈ B, compat2 s t → s = t
3263     ) → B.card ≤ 2
3264
3265 def tensDual : Set (W × W → ℝ≥0∞) :=
3266   {t | ∀ y ∈ lin Bool Bool, ∀ z ∈ lin Bool
3267     Bool, ∑ p, ∑ q, y p * t (p, q) * z q
3268     ≤ 1}
3269
3270 def sequential : Set (W × W → ℝ≥0∞) :=
3271   {t | ∃ (α β : ℝ≥0∞) (F : FFirst) (G :
3272     GFirst), α + β ≤ 1 ∧ t = α â¢ F.den
3273     + β â¢ G.den}
3274
3275 theorem seq_pent_le_two {t : W × W → ℝ≥0∞}
3276   (ht : t ∈ sequential) : ∑ pq ∈ pent, t
3277     pq ≤ 2
3278
3279 theorem seq_pent_tight : ∃ t ∈ sequential,
3280   ∑ pq ∈ pent, t pq = 2
3281
3282 theorem pentElt_pent_sum : ∑ pq ∈ pent,
3283   pentElt pq = 5 * 2-1
3284
3285 theorem tensDual_pent_le {t : W × W → ℝ≥0∞}
3286   (ht : t ∈ tensDual) :
3287     2 * ∑ pq ∈ pent, t pq ≤ 5
3288
3289 theorem sharp_bang_mixed_zero {z : ℕ × ℕ →
3290   ℝ≥0∞} (hz : z ∈ bang) (h01 : ∀ m, z m
3291     = 0 ∨ z m = 1)
3292   {i j : ℕ} (hi : 1 ≤ i) (hj : 1 ≤ j) :
3293     z (i, j) = 0
3294
3295 noncomputable def prom (x : Bool → ℝ≥0∞) :
3296   ℕ × ℕ → ℝ≥0∞ := fun m => x true ^ m.1
3297     * x false ^ m.2
3298
3299
3300

```

```

3301 def bang : Set (ℕ × ℕ → ℝ≥0∞) := orthI (
3302   orthI gens)
3303
3304 theorem prom_half_not_mix : prom half ∈
3305   bang ∧
3306   prom half ∉ Mix {z | z ∈ bang ∧ ∀ m, z
3307     m = 0 ∨ z m = 1}
3308
3309 def excl (s t : (X × Y) × (X' × Y')) :
3310   Prop := s ≠ t ∧ cp2 s t
3311
3312 theorem tD_eq_QSTAB : tD X Y X' Y' = QSTAB
3313   (excl (X := X) (Y := Y) (X' := X') (Y'
3314     := Y'))
3315
3316 theorem recipe_eq_STAB :
3317   Mix {z | z ∈ tD X Y X' Y' ∧ ∀ u, z u =
3318     0 ∨ z u = 1}
3319   = STAB (excl (X := X) (Y := Y) (X'
3320     := X') (Y' := Y'))
3321
3322 def QSTAB : Set (V → ℝ≥0∞) := {x | ∀ C,
3323   IsCliqueG adj C → ∑ v ∈ C, x v ≤ 1}
3324
3325 def STAB : Set (V → ℝ≥0∞) := Mix {x | ∃ I,
3326   IsStableG adj I ∧ x = ind I}
3327
3328 structure Hole (adj : V → V → Prop) (n : ℕ
3329   ) where
3330   h : ZMod n → V
3331   inj : Function.Injective h
3332   adj_iff : ∀ i j, i ≠ j → (adj (h i) (h j)
3333     ) ↔ cyc i j)
3334
3335 theorem berge_of_stab_eq_qstab (heq : STAB
3336   adj = QSTAB adj) (k : ℕ) (hk : 2 ≤ k)
3337   [NeZero (2 * k + 1)] : IsEmpty (Hole
3338     adj (2 * k + 1)) ∧ IsEmpty (
3339     Antihole adj (2 * k + 1))
3340
3341 theorem berge_of_recipe_eq_pcs
3342   (heq : Mix {z | z ∈ tD X Y X' Y' ∧ ∀ u
3343     , z u = 0 ∨ z u = 1} = tD X Y X' Y'
3344     ')
3345   (k : ℕ) (hk : 2 ≤ k) [NeZero (2 * k +
3346     1)] :
3347   IsEmpty (Hole (excl (X := X) (Y := Y)
3348     (X' := X') (Y' := Y')) (2 * k + 1))
3349   ∧
3350   IsEmpty (Antihole (excl (X := X) (Y
3351     := Y) (X' := X') (Y' := Y')) (2 *
3352     k + 1))
3353
3354
3355

```

B.11 Multiplicative-Additive Formulas

```

3356 inductive Fm where
3357   | base (n : ℕ)
3358   | neg (A : Fm)
3359   | tens (A B : Fm)
3360   | wth (A B : Fm)
3361
3362 def web : Fm → Type
3363   | base n => Fin n
3364   | neg A => web A
3365   | tens A B => web A × web B
3366   | wth A B => web A × web B
3367
3368 def P : (A : Fm) → Set (web A → ℝ≥0∞)
3369   | base n => flat (Fin n)
3370   | neg A => orth (P A)
3371   | tens A B => orth (orth {z | ∃ x ∈ P A,
3372     ∃ y ∈ P B, z = tvec x y})
3373   | wth A B => {z | (fun a => z (Sum.inl a
3374     )) ∈ P A ∧ (fun b => z (Sum.inr b)) ∈
3375     P B}
3376
3377 def coh : (A : Fm) → web A → web A → Prop
3378   | base _ => fun x y => x = y
3379   | neg A => fun x y => x = y ∨ ¬ coh A x
3380     y
3381   | tens A B => fun s t => coh A s.1 t.1 ∧
3382     coh B s.2 t.2
3383   | wth A B => fun s t => match s, t with
3384     | Sum.inl a, Sum.inl a' => coh A a a'
3385     | Sum.inr b, Sum.inr b' => coh B b b'
3386     | _, _ => True
3387
3388 theorem isPCS_P : ∀ A : Fm, IsPCS (P A)
3389
3390 theorem invariants : ∀ A : Fm,
3391   (∀ s : web A, Pi.single s 1 ∈ P A ∧ ∀
3392     x ∈ P A, x s ≤ 1) ∧
3393   (∀ s t : web A, s ≠ t → coh A s t → pr
3394     s t ∈ P A) ∧
3395   (∀ s t : web A, s ≠ t → ¬ coh A s t →
3396     ∀ x ∈ P A, x s + x t ≤ 1)
3397
3398 theorem single_mem_P (A : Fm) (s : web A)
3399   : Pi.single s 1 ∈ P A
3400
3401 theorem pair_mem_P (A : Fm) {s t : web A}
3402   (hst : s ≠ t) (h : coh A s t) : pr s t
3403   ∈ P A
3404
3405 theorem sharp_isClique (A : Fm) {z : web A
3406   → ℝ≥0∞} (hz : z ∈ P A)
3407
3408
3409
3410

```

```

3411      (h01 :  $\forall s, z s = 0 \vee z s = 1$ ) :  $\forall s t$ 
3412      ,  $z s = 1 \rightarrow z t = 1 \rightarrow \text{coh } A s t$ 
3413
3414 theorem recipe_subset_P (A : Fm) :
3415   Mix {z | z  $\in$  P A  $\wedge \forall s, z s = 0 \vee z s$ 
3416   = 1}  $\subseteq$  P A
3417
3418 theorem recipe_sharp_iff (A : Fm) {x : web
3419   A  $\rightarrow \mathbb{R}_{\geq 0\infty}$ } (hx :  $\forall s, x s = 0 \vee x s =$ 
3420   1) :
3421   x  $\in$  Mix {z | z  $\in$  P A  $\wedge \forall s, z s = 0 \vee$ 
3422   z s = 1}  $\leftrightarrow$  x  $\in$  P A
3423
3424 theorem Mix_subset_orth {S T : Set (E  $\rightarrow \mathbb{R}_{\geq}$ 
3425   0 $\infty$ )} (h : S  $\subseteq$  orth T) : Mix S  $\subseteq$  orth T
3426
3427 theorem P_lolli_base (n m :  $\mathbb{N}$ ) : P (lolli
3428   (base n) (base m)) = lin (Fin n) (Fin m)
3429   )
3430
3431 def second (n m n' m' :  $\mathbb{N}$ ) : Fm := neg (
3432   tens (lolli (base n) (base m)) (lolli (
3433   base n') (base m'))))
3434
3435 theorem P_second (n m n' m' :  $\mathbb{N}$ ) :
3436   P (second n m n' m') = tD (Fin n) (Fin
3437   m) (Fin n') (Fin m')
3438
3439 theorem recipe_first_order (n m :  $\mathbb{N}$ ) :
3440   P (lolli (base n) (base m)) =
3441   ConstructiveProb.Mix (Set.range
3442   (det : (Fin n  $\rightarrow$  Option (Fin m))  $\rightarrow$ 
3443   Fin n  $\times$  Fin m  $\rightarrow \mathbb{R}_{\geq 0\infty}$ ))
3444
3445 theorem recipe_second_order {n m n' m' :  $\mathbb{N}$ 
3446   } (hn : 0 < n) (hn' : 0 < n') :
3447   ConstructiveProb.Mix {z | z  $\in$  P (
3448   second n m n' m')  $\wedge \forall u, z u = 0 \vee$ 
3449   z u = 1}
3450   = sequentialG (Fin n) (Fin m) (Fin n
3451   ') (Fin m')
3452
3453 theorem recipe_second_order_ssubset {n m n
3454   ' m' :  $\mathbb{N}$ } (hn : 2  $\leq$  n) (hm : 2  $\leq$  m) (hn
3455   ' : 2  $\leq$  n')
3456   (hm' : 2  $\leq$  m') :
3457   ConstructiveProb.Mix {z | z  $\in$  P (
3458   second n m n' m')  $\wedge \forall u, z u = 0 \vee$ 
3459   z u = 1}
3460    $\hat{=}$  P (second n m n' m')
3461
3462 abbrev S2 : Fm := second 2 2 2 2
3463
3464 abbrev TT : Fm := neg (tens S2 S2)
3465

```

```

3466 noncomputable def shannonF : Finset (web (
3467   tens S2 S2)) := shannon.image fun u =>
3468   ( $\bar{I}$  u.1,  $\bar{I}$  u.2)
3469
3470 theorem shannonF_clique :  $\forall u \in$  shannonF,
3471    $\forall v \in$  shannonF, coh TT u v
3472
3473 theorem shannonF_not_mem : (fun u => if u
3474    $\in$  shannonF then (1 :  $\mathbb{R}_{\geq 0\infty}$ ) else 0)  $\notin$  P
3475   TT
3476
3477
3478
3479
3480

```

B.12 Quantum Event Logic

```

3481 def vec : Fin 18  $\rightarrow$  Fin 4  $\rightarrow \mathbb{Z}$  := ![
3482   ![-1, 1, 1, 1], ![0, 0, 0, 1], ![0, 0,
3483   1, 0], ![0, 0, 1, 1], ![0, 1, -1, 0],
3484   ![0, 1, 0, -1],
3485   ![0, 1, 0, 0], ![1, -1, -1, 1], ![1, -1,
3486   0, 0], ![1, -1, 1, -1], ![1, 0, -1,
3487   0], ![1, 0, 0, -1],
3488   ![1, 0, 0, 1], ![1, 0, 1, 0], ![1, 1,
3489   -1, 1], ![1, 1, 0, 0], ![1, 1, 1,
3490   -1], ![1, 1, 1, 1]]
3491
3492 def bas : Fin 9  $\rightarrow$  Fin 4  $\rightarrow$  Fin 18 := ![
3493   ![1, 2, 15, 8], ![1, 6, 13, 10], ![9, 7,
3494   15, 3], ![9, 17, 10, 5], ![2, 6, 12,
3495   11],
3496   ![7, 17, 11, 4], ![14, 16, 8, 3], ![14,
3497   0, 13, 5], ![16, 0, 12, 4]]
3498
3499 def dot (u v : Fin 4  $\rightarrow \mathbb{Z}$ ) :  $\mathbb{Z}$  :=  $\sum k, u k$ 
3500   * v k
3501
3502 theorem basis_orth :  $\forall b : \text{Fin } 9, \forall i j :$ 
3503   Fin 4,  $i \neq j \rightarrow \text{dot (vec (bas } b \ i)) (vec$ 
3504   (bas b j)) = 0
3505
3506 theorem occ_two :  $\forall v : \text{Fin } 18,$ 
3507   (univ.filter (fun p : Fin 9  $\times$  Fin 4 =>
3508   bas p.1 p.2 = v)).card = 2
3509
3510 theorem no_ks :  $\neg \exists f : \text{Fin } 18 \rightarrow \text{Bool}, \forall b$ 
3511   , (univ.filter (fun i => f (bas b i))).
3512   card = 1
3513
3514 def certain : Set (Fin 18  $\rightarrow \mathbb{R}_{\geq 0\infty}$ ) :=
3515   {s | IsSharpVec s  $\wedge \forall b, \sum i, s$  (bas b i)
3516   = 1}
3517
3518 theorem no_certain_state : certain =  $\emptyset$ 
3519
3520

```

```

3521 theorem Mix_certain_empty : Mix certain =
3522   0
3523
3524 theorem parseval (I : Fin 4 → ℝ) (b : Fin
3525   9) :
3526   ∑ i, (∑ k, I k * vec (bas b i) k) ^ 2
3527   / ∑ k, ((vec (bas b i) k : ℝ)) ^ 2
3528   = ∑ k, I k ^ 2
3529
3530 def states : Set (Fin 18 → ℝ≥0∞) := {x | ∀
3531   b, ∑ i, x (bas b i) = 1}
3532
3533 noncomputable def born (I : Fin 4 → ℝ) :
3534   Fin 18 → ℝ≥0∞ := fun v =>
3535   ENNReal.ofReal ((∑ k, I k * vec v k) ^
3536     2 / ∑ k, ((vec v k : ℝ)) ^ 2)
3537
3538 theorem born_mem {I : Fin 4 → ℝ} (hI : ∑
3539   k, I k ^ 2 = 1) : born I ∈ states
3540
3541 theorem born_not_mix {I : Fin 4 → ℝ} (hI
3542   : ∑ k, I k ^ 2 = 1) :
3543   born I ∈ states ∧ born I ∉ Mix
3544   certain
3545

```

B.13 Finite Frames without Choice

```

3546
3547 def JoinPrime (j : α) : Prop := ∀ x y : α,
3548   j ≤ x ⊔ y → j ≤ x ∨ j ≤ y
3549
3550
3551 theorem exists_joinPrime_aux : ∀ (n : ℕ) (
3552   a b : α), (below a).card = n → ¬ a ≤ b
3553   →
3554   ∃ j, j ≤ a ∧ ¬ j ≤ b ∧ JoinPrime j
3555
3556 theorem exists_sharp_separating_fin [
3557   BoundedOrder α] {a b : α} (hab : ¬ a ≤
3558   b) :
3559   ∃ v : α → ℕ, (∀ x, v x = 0 ∨ v x = 1)
3560   ∧ Monotone v ∧ v ⊥ = 0 ∧ v ⊤ = 1 ∧
3561   (∀ x y, v x + v y = v (x ⊔ y) + v (x
3562     ⊓ y)) ∧ v a = 1 ∧ v b = 0
3563
3564 theorem em_of_sharp_complement_fin [
3565   BoundedOrder α] (a c : α)
3566   (h : ∀ v : α → ℕ, (∀ x, v x = 0 ∨ v x
3567     = 1) → Monotone v → v ⊥ = 0 → v ⊤ =
3568     1 →
3569     (∀ x y, v x + v y = v (x ⊔ y) + v (x
3570       ⊓ y)) → v a + v c = 1) :
3571   a ⊔ c = ⊤
3572

```

B.14 Choice-Free Finite Results

```

3573
3574
3575

```

```

3576 inductive Op | bot | halts | top
3577   deriving DecidableEq
3578
3579 def neg : Op → Op
3580   | bot => top
3581   | halts => bot
3582   | top => bot
3583
3584 theorem neg_spec : ∀ o o' : Op, (inf o' o
3585   = bot) ↔ le o' (neg o) = true
3586
3587 theorem neg_halts : neg halts = bot
3588
3589 theorem frontier_iff : ∀ x : Bool,
3590   frontier x = true ↔ x = false
3591
3592 theorem slack_eq (wT wF : Nat) :
3593   val wT wF top = val wT wF halts + val
3594   wT wF (neg halts) + wt (frontier
3595     false) wF
3596
3597 def ext (ops : List (Fin n → Bool)) (U :
3598   Fin n → Bool) (x : Fin n) : Bool :=
3599   ops.any fun 0 => 0 x && allFin n (fun y
3600     => !(0 y && U y))
3601
3602 def bd (ops : List (Fin n → Bool)) (U :
3603   Fin n → Bool) (x : Fin n) : Bool :=
3604   !U x && !ext ops U x
3605
3606 theorem sub_ext {ops : List (Fin n → Bool)
3607   } {U : Fin n → Bool} {0 : Fin n → Bool}
3608   (h0 : 0 ∈ ops) (hd : ∀ y, (!(0 y && U
3609     y)) = true) {x : Fin n} (hx : 0 x =
3610     true) :
3611   ext ops U x = true
3612
3613 theorem slack_eq_boundary_fin (ops : List
3614   (Fin n → Bool)) (U : Fin n → Bool) (μ :
3615   Fin n → Nat) :
3616   fs n μ = fs n (fun x => wt (U x) (μ x)
3617     ) + fs n (fun x => wt (ext ops U x)
3618     (μ x)) +
3619   fs n (fun x => wt (bd ops U x) (μ x)
3620     )
3621
3622 def ksum (L : Nat) : List (List Bool) →
3623   Nat
3624   | [] => 0
3625   | w :: S => 2 ^ (L - w.length) + ksum L
3626     S
3627

```

```

3631 theorem kraft :  $\forall (L : \text{Nat}) (S : \text{List } ($ 
3632    $\text{List Bool}))$ ,  $\text{PF } S \rightarrow (\forall w \in S, w.\text{length}$ 
3633    $\leq L) \rightarrow$ 
3634    $\text{ksum } L \ S \leq 2^L$ 
3635
3636 structure Row (n : Nat) where
3637   c : Fin n  $\rightarrow$  Int
3638   s : Bool
3639
3640 inductive Der {n : Nat} (rows : List (Row
3641   n)) : Row n  $\rightarrow$  Prop
3642   | base {r : Row n} : r  $\in$  rows  $\rightarrow$  Der rows
3643     r
3644   | comb {r1 r2 : Row n} (p q : Int) :  $0 \leq$ 
3645     p  $\rightarrow 0 \leq q \rightarrow$  Der rows r1  $\rightarrow$  Der rows r
3646     2  $\rightarrow$ 
3647     Der rows (comb p q r1 r2)
3648
3649 theorem fm :  $\forall (n : \text{Nat}) (rows : \text{List } (\text{Row}$ 
3650   n)),  $(\exists w, \forall r \in \text{rows}, \text{Sat } r \ w) \vee$  Der
3651   rows (zeroS n)
3652
3653 theorem complete (hd :  $0 < d$ )
3654   (hlaw :  $\forall (c : \text{Fin } m \rightarrow \text{Int}) (k : \text{Int}),$ 
3655      $(\forall j, k \leq \text{dot } m \ c \ (S \ j)) \rightarrow k * d \leq$ 
3656      $\text{dot } m \ c \ y)$  :
3657    $\exists (\text{lam} : \text{Fin } N \rightarrow \text{Int}) (t : \text{Int}), 0 < t$ 
3658      $\wedge (\forall j, 0 \leq \text{lam } j) \wedge$ 
3659      $\text{dot } N \ (\text{fun } _ \Rightarrow 1) \ \text{lam} = t \wedge \forall e, d$ 
3660      $* \text{dot } N \ \text{lam} \ (\text{fun } j \Rightarrow S \ j \ e) = t$ 
3661      $* y \ e$ 
3662
3663 theorem sound (hd :  $0 < d$ ) (lam : Fin N  $\rightarrow$ 
3664   Int) (t : Int) (ht :  $0 < t$ ) (hl :  $\forall j,$ 
3665      $0 \leq \text{lam } j$ )
3666   (hsum :  $\text{dot } N \ (\text{fun } _ \Rightarrow 1) \ \text{lam} = t$ ) (
3667     hx :  $\forall e, d * \text{dot } N \ \text{lam} \ (\text{fun } j \Rightarrow S$ 
3668       j e) = t * y e)
3669   (c : Fin m  $\rightarrow$  Int) (k : Int) (hc :  $\forall j,$ 
3670      $k \leq \text{dot } m \ c \ (S \ j))$  :  $k * d \leq \text{dot } m$ 
3671     c y
3672
3673 structure EOD (A : Nat  $\rightarrow$  Nat  $\rightarrow$  Bool) where
3674   N : Nat
3675   lt : Bool
3676   eq : Bool
3677   gt : Bool
3678   spec :  $\forall m \ m', N \leq m \rightarrow N \leq m' \rightarrow$ 
3679     A m m' = if m < m' then lt else if m =
3680     m' then eq else gt
3681
3682 def L {A : Nat  $\rightarrow$  Nat  $\rightarrow$  Bool} (w : EOD A) :
3683   Bool := w.lt
3684
3685

```

```

-- The iterated limit with `m  $\rightarrow \infty$ ` first.
-/
def R {A : Nat  $\rightarrow$  Nat  $\rightarrow$  Bool} (w : EOD A) :
  Bool := w.gt

def R {A : Nat  $\rightarrow$  Nat  $\rightarrow$  Bool} (w : EOD A) :
  Bool := w.gt

theorem L_modular {A B} (a : EOD A) (b :
  EOD B) :
  val (L a) + val (L b) = val (L (union
    a b)) + val (L (inter a b))

theorem rect_L {S T} (s : EC S) (t : EC T)
  : L (rect s t) = (s.v && t.v)

theorem rect_R {S T} (s : EC S) (t : EC T)
  : R (rect s t) = (s.v && t.v)

theorem products_differ : L tri  $\neq$  R tri

```

C The Aczél Development

This appendix sketches the from-scratch construction behind the product-rule half of Theorem 30. The target shape is Aczél's associativity theorem [Aczél 1966]: a suitably regular associative $F : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ regroups to multiplication, $G(Fxy) = G(x) \cdot G(y)$ for all x, y and a strictly monotone G . mathlib has no associativity representation, no Hölder-style embedding of ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity).

Hypotheses. A Scale bundles: associativity of F , *divisibility* $\forall c \exists x, Fxx = c$, positivity on the cone $x < Fxc$, joint order-preservation, joint continuity, a unit u , and identity at $-\infty$, meaning $Fxz \rightarrow x$ as $z \rightarrow -\infty$. Order-preservation is required by the additivity proof below. Divisibility yields the 2^n -th roots used in the dyadic construction below. Without it the generator is not definable. It is assumed directly in the bundle, though its interval form follows from the intermediate value theorem (exists_diag_eq). Non-vacuity: *log-sum-exp*, $Fxy = \log(e^x + e^y)$ with $u = 0$, satisfies every field of the bundle (logSumExpScale), with intended generator globally exp.

Construction. Write $a^{(n)}$ for n right-associated copies of a under F (Fpow). Associativity alone makes $n \mapsto a^{(n)}$ a semigroup morphism from $(\mathbb{N}, +)$. Commutativity is not assumed, and follows instead as a corollary of the generator. Unboundedness of the iterates is derived: were they bounded, monotone convergence would give a fixed point $FLa = L$, contradicting positivity, and unboundedness yields the Archimedean property. Divisibility yields a tower of 2^n -th roots $u \succ u^{1/2} \succ u^{1/4} \succ \dots$ (droot), where 2^l copies of

the finer root compose under F to one copy of the coarser root. Counting the number of copies of the n -th root not exceeding b (dcount, well defined by the Archimedean property) gives dyadic approximants $g_n(b) = \text{dcount}(b, n)/2^n$. The refinement inequality $2N + 1 \leq N' \leq 2N + 2$, where N and N' are the counts at consecutive refinement levels n and $n + 1$, makes (g_n) monotone and Cauchy, and the generator is its limit, $g(b) = \bigsqcup_n g_n(b)$.

Results (all sorry-free, standard axioms). On the cone $\{x \mid u \leq x\}$ we prove three things. Normalization: $g(u) = 1$. Additivity: $g(Fx y) = g(x) + g(y)$, since $\text{dcount } x + \text{dcount } y + 1 \leq \text{dcount}(Fx y) \leq \text{dcount } x + \text{dcount } y + 3$ and the 2^{-n} error term vanishes. And strict monotonicity, proved without any continuity of g : a root inserted between $x < y$ contributes a full extra block of 2^{n-M} copies, M being the refinement level of the

inserted root. Exponentiating gives the multiplicative form, a strictly monotone G with $G(Fx y) = G(x)G(y)$ on the cone (exists_mul_generator), the conclusion of Aczél's theorem restricted to the cone (aczelStatement_cone). The reorientation $\bar{G} = \exp(-g)$ gives the $[0, 1]$ -valued form used in Section 6.

Remaining gap. The constructed g is discontinuous at the unit (it is 0 below u), so matching the classical statement verbatim needs the off-cone extension (a group completion) and continuity of the extension. This is open. It is not vacuous: the log-sum-exp example admits a global continuous generator (hasOrderedGenerator_logSumExp). Of Hölder's classical hypotheses, the Archimedean property is proved, and is what enables the embedding into $\mathbb{R}$. The interval form of divisibility follows from the intermediate value theorem. Only the embedding's global regularity remains.